Option-Implied Event Contracts: A Complete Framework for Derivatives on Prediction Markets

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Abstract

This paper develops a comprehensive mathematical framework for pricing and hedging derivative contracts written on prediction market event futures. We introduce *Option-Implied Event Contracts* (OIECs), a new class of derivatives that treats the price of a prediction market contract as a bounded underlying asset evolving on $[0, 1]$. Using the Jacobi diffusion process as a natural model for bounded probability dynamics, we derive closed-form pricing formulas for European-style options, establish put-call parity relations, develop a complete Greeks calculus, and prove market completeness results. We extend the framework to include confidence-weighted event digitals, belief variance swaps with model-free replication, American-style early exercise features, and correlated multi-event derivatives. The paper provides explicit calibration procedures, a worked numerical example using historical prediction market data, margin methodology for exchange clearing, a prototype product specification, and market-making economics with illustrative dollar figures. We further develop a model-free Belief Volatility Index (BVIX), path-dependent regime contracts, a probability-weighted portfolio theory with state-dependent efficient frontier, Jacobi bonding curves for automated market structure, a Greeks-bounded market-making protocol with deterministic loss bounds, self-referential prediction derivatives with fixed-point pricing, and a maximum-entropy nonparametric pricing framework. Our framework bridges the gap between prediction markets and traditional derivatives infrastructure, enabling institutional participation through familiar risk-management tools.

Keywords: Prediction markets, event derivatives, bounded diffusions, Jacobi process, option pricing, market completeness, belief volatility index, bonding curves

JEL Classification: G13, G14, D84

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1 Introduction

Think about the last time you made a bet, not necessarily with money, but with your choices. Perhaps you grabbed an umbrella before leaving home, left early for a flight, or moved to cash before an election. Each of these decisions involved uncertainty. You acted not because you knew what would happen, but because you had a sense of what might happen. Carrying an umbrella is not a forecast of rain; it is a hedge against the chance of getting wet.

Prediction markets formalize that instinct. A contract on "Fed raises rates in March" trading at 0.73 encodes the collective view that the event is roughly three-quarters likely, and that view tends to be well-calibrated. Platforms like Polymarket have demonstrated this across elections, monetary policy, and macroeconomic indicators.

What prediction markets do not support is any expression of a view on *dynamics*. A trader who thinks the election is 50/50 can buy at 0.50. But a trader who expects the probability to stay near 0.50 for three months and then swing violently in the final week cannot do anything with that belief. There is no straddle, no vol surface, no term structure of uncertainty. The entire toolkit that lets equity and rates markets function as hedging and risk-transfer venues is absent.

This matters for practical reasons. Market makers on event futures carry inventory risk they cannot hedge with options. Institutions that would otherwise participate lack the Greeks-based risk management they require. And without a volatility surface, there is no market-implied measure of how uncertain the probability itself is. The underlying event future is there; the derivative layer that would make it useful to a broader set of participants is not.

This paper constructs that layer. The idea is simple: treat the prediction market price $P_t \in [0, 1]$ as a stochastic underlying and price derivatives on it. The Jacobi diffusion, a process confined to the unit interval with volatility proportional to $\sqrt{P(1-P)}$, provides the dynamics. From there, the standard machinery applies. We get European and American options, put-call parity, delta hedging, a pricing PDE, and market completeness. We also get objects that are new to this setting: belief variance swaps that isolate exposure to probability volatility, confidence-weighted digitals that pay on joint market-assessment-and-outcome events, and model-free variance replication through a static option strip.

The framework is called Option-Implied Event Contracts (OIECs). The approach adapts classical option pricing to bounded state spaces. The usual Black-Scholes model assumes geometric Brownian motion, which can wander freely across positive real values. Probabilities must remain between 0 and 1 and eventually resolve to a binary outcome. The Jacobi diffusion provides a natural model. Building on this process, we derive closed-form pricing results, extend put-call parity to bounded underlyings, define the associated Greeks, and prove market completeness via the fundamental theorems of asset pricing (Harrison and Kreps, 1979; Harrison and Pliska, 1981; Delbaen and Schachermayer, 1994). The model-free replication of belief variance swaps extends the celebrated log-contract result of Neuberger (1994) and the general variance swap theory of Carr and Lee (2009); Demeterfi et al. (1999) to the bounded domain $[0, 1]$.

Beyond the mathematical results, this paper addresses questions an exchange would need answered before listing these products. What does the product specification look like? How should margin be set? What are the economics for a market maker? We provide concrete answers.

The remainder of this paper is organized as follows. Section 2 establishes the mathematical foundations, including the no-arbitrage framework and the Jacobi diffusion model for bounded probability dynamics. Section 3 formally defines the OIEC contract family. Section 4 derives pricing formulas and establishes put-call parity. Section 5 develops the Greeks calculus and hedging strategies. Section 6 proves market completeness results. Section 7 extends the framework to American options, correlated multi-event derivatives, and exotic payoffs. Section 8 presents numerical methods. Section 9 discusses calibration to market data. Section 10 provides a worked

numerical example. Section 11 gives a prototype product specification. Section 12 develops margin methodology. Section 13 analyzes market-making economics. Section 21 illustrates economic applications. Section 22 addresses regulatory and implementation considerations.

Part II of the paper develops advanced extensions. Section 14 constructs the Belief Volatility Index. Section 15 introduces path-dependent regime contracts. Section 16 develops probability-weighted portfolio theory. Section 17 proposes Jacobi bonding curves. Section 18 presents automated market making with provable loss bounds. Section 19 develops self-referential prediction derivatives. Section 20 derives a maximum-entropy nonparametric pricing framework. Section 23 concludes.

2 Model Setup

2.1 Probability Space and Filtration

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space supporting a standard Brownian motion $(W_t)_{t\geq 0}$. Fix a finite horizon $T > 0$ and let $(\mathcal{F}_t)_{t\in[0,T]}$ be the augmented filtration generated by W.

2.2 Event Futures

Consider a binary event $E \in \mathcal{F}_T$ with maturity T. We interpret E as a measurable event whose occurrence is revealed at time T. Examples include:

- $E = \{\text{Candidate } A \text{ wins the 2028 presidential election}\}$
- $E = \{\text{Fed raises rates by at least 25 bps at March meeting}\}$
- $E = \{\text{Inflation exceeds 3\% in Q4 2026}\}$

Definition 2.1 (Event Future)**.** An *event future* on E is a cash-settled contract that pays

$$X_T := \mathbf{1}_{\{E\}} = \begin{cases} 1 & \text{if } E \text{ occurs,} \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

at maturity T. Let P_t denote the time-t price of this event future, quoted in normalized units so that $P_t \in [0, 1]$ for all $t \in [0, T]$.

2.3 Money Market Account

Let $(r_t)_{t\in[0,T]}$ be a deterministic short rate process. The money market account is

$$B_t := \exp\left(\int_0^t r_s \, \mathrm{d}s\right). \tag{2}$$

For most of this paper, we work with constant $r \geq 0$, so $B_t = e^{rt}$.

2.4 No-Arbitrage and Risk-Neutral Pricing

Assumption 2.2 (No-Arbitrage)**.** There exists an equivalent probability measure $\mathbb{Q} \sim \mathbb{P}$ such that the discounted price process $\tilde{P}_t := B_t^{-1} P_t$ is a $\mathbb{Q}$-martingale.

Under Assumption 2.2, the event future price satisfies

$$P_t = B_t \, \mathbb{E}^{\mathbb{Q}} \left[B_T^{-1} X_T \,\middle|\, \mathcal{F}_t \right] = e^{-r(T-t)} \, \mathbb{Q}(E \mid \mathcal{F}_t). \tag{3}$$

Thus P_t is (up to discounting) the $\mathbb{Q}$-conditional probability of E.

Proposition 2.3 (No-Arbitrage Bounds)**.** *Under Assumption 2.2, the event future price satisfies*

$$0 \leq P_t \leq e^{-r(T-t)} \leq 1 \tag{4}$$

for all $t \in [0, T]$.

Proof. Since $X_T \in \{0, 1\}$ and $\mathbb{Q}(E \mid \mathcal{F}_t) \in [0, 1]$, the bound follows directly from (3). □

Proposition 2.4 (EMMs Imply No Arbitrage)**.** *If there exists* $\mathbb{Q} \sim \mathbb{P}$ *such that* $\tilde{P}_t$ *is a* $\mathbb{Q}$*-martingale, then the market admits no arbitrage among self-financing strategies with finite initial wealth.*

Proof. Let $\varphi = (\varphi_t^0, \varphi_t^1)_{t\in[0,T]}$ be a self-financing strategy trading the money market and event future. The portfolio value is $V_t(\varphi) = \varphi_t^0 B_t + \varphi_t^1 P_t$, and the discounted value is

$$\tilde{V}_t(\varphi) := B_t^{-1} V_t(\varphi) = \tilde{V}_0(\varphi) + \int_0^t \varphi_s^1 \, \mathrm{d}\tilde{P}_s.$$

Since $\tilde{P}$ is a $\mathbb{Q}$-martingale, so is $\tilde{V}(\varphi)$ under standard integrability conditions. If φ were an arbitrage with $V_0 \leq 0$, $V_T \geq 0$ a.s., and $\mathbb{P}(V_T > 0) > 0$, then equivalence $\mathbb{Q} \sim \mathbb{P}$ implies $\mathbb{Q}(V_T > 0) > 0$, so $\mathbb{E}^{\mathbb{Q}}[\tilde{V}_T] > 0 = \tilde{V}_0$, contradicting the martingale property. □

2.5 The Jacobi Diffusion Model

To price derivatives on P_t, we require a tractable dynamic model that respects the $[0, 1]$ bounds. The *Jacobi diffusion* (also called the Wright–Fisher diffusion in population genetics) is a natural choice.

Definition 2.5 (Jacobi Diffusion)**.** The Jacobi diffusion on $[0, 1]$ with parameters $\alpha, \beta > 0$ and volatility scaling $\sigma > 0$ satisfies

$$\mathrm{d}P_t = \kappa(\theta - P_t)\,\mathrm{d}t + \sigma\sqrt{P_t(1 - P_t)}\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{5}$$

where $\kappa > 0$ is the mean-reversion speed, $\theta \in (0, 1)$ is the long-run mean, and $(W_t^{\mathbb{Q}})$ is a standard Brownian motion under $\mathbb{Q}$.

The drift term $\kappa(\theta - P_t)$ pulls the process toward θ, while the diffusion term $\sigma\sqrt{P_t(1 - P_t)}$ vanishes at the boundaries, ensuring $P_t \in [0, 1]$ for all t.

Remark 2.6 (Alternative Parameterization)**.** The Jacobi parameters α, β relate to κ, θ via

$$\alpha = \frac{2\kappa\theta}{\sigma^2}, \quad \beta = \frac{2\kappa(1 - \theta)}{\sigma^2}.$$

When $\alpha, \beta > 1$ (the strict Feller condition), the boundaries 0 and 1 are unattainable in finite time. When $\alpha, \beta \leq 1$, the boundaries become attainable.

Under the risk-neutral measure, the discounted event future must be a martingale. For the Jacobi process (5) to satisfy this, we require a driftless specification:

$$\mathrm{d}P_t = \sigma\sqrt{P_t(1 - P_t)}\,\mathrm{d}W_t^{\mathbb{Q}}. \tag{6}$$

Proposition 2.7 (Jacobi Martingale)**.** *The process* (P_t) *defined by* (6) *with* $P_0 \in (0, 1)$ *is a* $[0, 1]$*-valued martingale under* $\mathbb{Q}$.

Proof. The drift is zero, so P_t is a local martingale. Since $P_t \in [0, 1]$, the process is bounded, hence a true martingale. □

Remark 2.8 (Boundary Attainability in the Martingale Case)**.** Setting $\kappa = 0$ in Remark 2.6 yields $\alpha = \beta = 1$, which lies exactly on the Feller boundary. In this regime, the boundaries $\{0, 1\}$ are *attainable* in finite time: the process can reach 0 or 1. This is in fact desirable for prediction markets, since event futures must eventually resolve to a binary outcome. The martingale Jacobi process thus naturally accommodates the terminal resolution of the event. However, for option pricing purposes with $\tau < T$, the continuous-path behavior on $(0, 1)$ is the relevant regime.

Remark 2.9 (Physical Measure Dynamics and Girsanov Change of Measure)**.** Under the physical measure $\mathbb{P}$, prediction market prices exhibit drift reflecting information arrival and sentiment shifts. Suppose the $\mathbb{P}$-dynamics are given by the general Jacobi form (5) with drift $\mu(P_t, t)$. The Girsanov theorem provides a change of measure via the Radon–Nikodym derivative

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \exp\left(-\int_0^t \lambda_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^t \lambda_s^2 \,\mathrm{d}s\right),$$

where the market price of risk is

$$\lambda_t = \frac{\mu(P_t, t)}{\sigma\sqrt{P_t(1 - P_t)}}.$$

Under $\mathbb{Q}$, the drift vanishes and the process satisfies (6). This change of measure is well-defined provided λ_t satisfies the Novikov condition. In practice, λ_t captures the compensation required by risk-neutral investors for bearing event uncertainty.

Remark 2.10 (Jump Extensions)**.** Prediction market prices jump on discrete news events: debate outcomes, court rulings, economic data releases. The continuous Jacobi model captures smooth information arrival but not these shocks. A natural extension adds a compound Poisson component:

$$\mathrm{d}P_t = \sigma\sqrt{P_t(1 - P_t)}\,\mathrm{d}W_t^{\mathbb{Q}} + J_t \,\mathrm{d}N_t, \tag{7}$$

where N_t is a Poisson process with intensity λ and J_t is a random jump size constrained so that P_t remains in $[0, 1]$. For instance, one may model J_t as a truncated normal or a beta-distributed shock scaled to keep the post-jump value in the unit interval. The continuous framework developed in the remainder of this paper applies between jumps; the jump component would modify option prices through additional tail risk and would break the completeness result (Theorem 6.1), requiring additional traded instruments to span the jump risk. A full treatment of the jump-diffusion Jacobi model is beyond the present scope, but the Greeks, PDE structure, and replication results extend with standard modifications (see, e.g., Merton (1973) for the analogous extension in the Black–Scholes setting). For contracts with τ sufficiently far from known news dates, the continuous model is a reasonable first approximation.

2.6 Transition Density of the Jacobi Process

The transition density of the martingale Jacobi process (6) is expressible in terms of Jacobi polynomials. Let $p(t, x; T, y)$ denote the density of $P_T = y$ given $P_t = x$.

Theorem 2.11 (Jacobi Transition Density)**.** *For the martingale Jacobi process* (6)*, the transition density on* $(0, 1)$ *is*

$$p(t, x; T, y) = \sum_{n=0}^{\infty} c_n(x) c_n(y) w(y) e^{-\lambda_n (T-t)}, \tag{8}$$

where:

- $w(y) = y^{\alpha-1}(1 - y)^{\beta-1}/B(\alpha, \beta)$ *is the* $\mathrm{Beta}(\alpha, \beta)$ *stationary density,*
- $c_n(y)$ *are orthonormal polynomials with respect to* w*,*

- $\lambda_n = \frac{\sigma^2}{2} n(n + \alpha + \beta - 1)$ *are the eigenvalues.*

For the martingale case ($\kappa = 0$), we have $\alpha = \beta = 1$, $w(y) = 1$ (uniform), and c_n are normalized Legendre polynomials $\tilde{P}_n(2y-1)$ with normalization $\tilde{P}_n = \sqrt{2n+1}\, P_n$, where P_n are the standard Legendre polynomials. The eigenvalues simplify to $\lambda_n = \frac{\sigma^2}{2} n(n+1)$.

Proof. This is a standard result for one-dimensional diffusions; see Karlin and Taylor (1981). The generator of (6) is $\mathcal{L}f(x) = \frac{\sigma^2}{2} x(1-x) f''(x)$. In the martingale case, the Sturm–Liouville problem $\mathcal{L}c_n = -\lambda_n c_n$ with respect to the uniform weight on $(0, 1)$ yields Legendre polynomial eigenfunctions. □

2.7 Terminal Distribution at Event Maturity

At $t = T$, the event E is revealed and P_T must equal $X_T \in \{0, 1\}$. This terminal condition requires the continuous process to collapse to the realized outcome. We address this through a separation of time scales.

Assumption 2.12 (Option Expiry Before Event Maturity)**.** All OIEC contracts have option maturity $\tau \leq T$. For $\tau < T$, the underlying P_τ is continuous on $(0, 1)$ and evolves according to the Jacobi dynamics (6). At $\tau = T$, claims settle based on the event realization X_T.

Remark 2.13 (Terminal Pinning and Information Arrival)**.** The transition from continuous Jacobi dynamics to binary resolution at T can be understood in two complementary ways. First, as $t \to T^-$, information arrival accelerates, and the effective volatility of the probability process increases, driving P_t toward 0 or 1. The Jacobi model with $\alpha = \beta = 1$ is consistent with this: the boundaries are attainable (Remark 2.8), so the process can reach $\{0, 1\}$ in finite time. Second, one may model a thin pre-terminal layer $(T - \varepsilon, T)$ during which a final information shock pins P_T to the realized outcome. For pricing options with τ bounded away from T, the distinction is immaterial: the Jacobi transition density applies on the continuous interior. For options with τ very close to T, a more detailed model of terminal information arrival—such as a Brownian bridge to the binary outcome or a hazard-rate model for decisive information shocks—may improve pricing accuracy. We leave this extension to future work.

3 Definition of Option-Implied Event Contracts

We now formally define the family of OIEC instruments.

3.1 European Probability Options

Definition 3.1 (European Probability Call and Put)**.** Fix option maturity $\tau \leq T$ and strike $K \in [0, 1]$. The *European probability call* has payoff

$$C^{\text{prob}}_{\tau,K} := (P_\tau - K)^+ = \max(P_\tau - K, 0), \tag{9}$$

and the *European probability put* has payoff

$$\Pi^{\text{prob}}_{\tau,K} := (K - P_\tau)^+ = \max(K - P_\tau, 0). \tag{10}$$

These are standard vanilla options but with the key distinction that the underlying P_τ is bounded in $[0, 1]$.

3.2 Digital (Binary) Probability Options

Definition 3.2 (Digital Probability Options)**.** The *cash-or-nothing call* pays $\mathbf{1}_{\{P_\tau \geq K\}}$, and the *asset-or-nothing call* pays $P_\tau \cdot \mathbf{1}_{\{P_\tau \geq K\}}$. Analogous puts pay on $\{P_\tau \leq K\}$.

3.3 Confidence-Weighted Event Digitals

The following contracts are unique to the OIEC framework, explicitly linking the terminal event realization with pre-event probability assessments.

Definition 3.3 (Confidence-Weighted Event Digital)**.** Fix $K \in [0, 1]$. The *upside confidence digital* pays

$$D_K^{\text{up}} := \mathbf{1}_{\{E\}} \cdot \mathbf{1}_{\{P_{T^-} \geq K\}}, \tag{11}$$

rewarding scenarios where the event occurs *and* the market was confident (probability $\geq K$) just before resolution. The *downside confidence digital* pays

$$D_K^{\text{down}} := \mathbf{1}_{\{E^c\}} \cdot \mathbf{1}_{\{P_{T^-} \leq K\}}, \tag{12}$$

rewarding scenarios where the event fails *and* the market was appropriately skeptical (probability $\leq K$).

Remark 3.4 (Economic Interpretation)**.** The confidence digitals provide exposure to the *joint* distribution of the event outcome and the market's pre-terminal assessment. Consider their use cases:

D_K^{up} with high K (e.g., $K = 0.8$): This pays when the event occurs and the market assigned high probability before resolution. It is a leveraged long bet on the event that rewards correct consensus. A trader who believes the market will converge to high confidence *and* the event will occur can purchase this at a discount relative to the event future itself.

D_K^{up} with low K (e.g., $K = 0.2$): This pays when the event occurs despite low market confidence—a genuine "surprise" contract. It is a bet on market miscalibration: the event happening when the market thought it would not. This is substantially cheaper and provides tail-event exposure.

D_K^{down} with high K (e.g., $K = 0.8$): This pays when the event *fails* despite the market being confident it would occur—another miscalibration bet, but on the opposite tail. It is valuable as a hedge against consensus overconfidence.

Together, the confidence digitals span a richer space of views than the event future alone, allowing traders to express opinions about market calibration quality.

3.4 Belief Variance Swaps

To isolate exposure to the volatility of market beliefs, we define contracts on the realized quadratic variation of the probability process.

Definition 3.5 (Belief Variance Swap)**.** Let $0 \leq t_0 < \tau \leq T$ and consider a partition $t_0 = s_0 < s_1 < \cdots < s_m = \tau$. The *realized belief variance* is

$$\text{RV}_{t_0,\tau}^{(P)} := \sum_{j=0}^{m-1} (P_{s_{j+1}} - P_{s_j})^2. \tag{13}$$

A *belief variance swap* with strike $K_{\text{var}} \geq 0$ has payoff

$$V_{t_0,\tau}^{(P)} := \text{RV}_{t_0,\tau}^{(P)} - K_{\text{var}}. \tag{14}$$

In the continuous limit, the quadratic variation of the Jacobi process is

$$[P]_{t_0,\tau} = \int_{t_0}^{\tau} \sigma^2 P_s (1 - P_s) \, \mathrm{d}s, \tag{15}$$

so the belief variance swap approximates exposure to integrated "probability variance."

3.5 Probability Straddles and Strangles

Definition 3.6 (Probability Straddle)**.** A *probability straddle* at strike K consists of a long call and a long put at the same strike:

$$\text{Straddle}_{\tau,K} = C^{\text{prob}}_{\tau,K} + \Pi^{\text{prob}}_{\tau,K} = |P_\tau - K|. \tag{16}$$

Definition 3.7 (Probability Strangle)**.** A *probability strangle* with strikes $K_1 < K_2$ is a long put at K_1 and a long call at K_2:

$$\text{Strangle}_{\tau,K_1,K_2} = \Pi^{\text{prob}}_{\tau,K_1} + C^{\text{prob}}_{\tau,K_2}. \tag{17}$$

4 Pricing Formulas

4.1 General Risk-Neutral Valuation

Under Assumption 2.2, the time-t price of any $\mathcal{F}_\tau$-measurable payoff H is

$$V_t(H) = e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t]. \tag{18}$$

4.2 European Probability Call Pricing

Theorem 4.1 (European Probability Call Price)**.** *Under the martingale Jacobi model* (6)*, the time-t price of a European probability call with maturity τ and strike $K \in (0,1)$ is*

$$c(t, P_t; \tau, K) = e^{-r(\tau-t)} \int_K^1 (y - K)\, p(t, P_t; \tau, y)\, \mathrm{d}y, \tag{19}$$

where $p(t, x; \tau, y)$ is the Jacobi transition density from Theorem 2.11.

Proof. By risk-neutral pricing,

$$c(t, P_t; \tau, K) = e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[(P_\tau - K)^+ \mid P_t] = e^{-r(\tau-t)} \int_K^1 (y - K)\, p(t, P_t; \tau, y)\, \mathrm{d}y.$$

□

Corollary 4.2 (Boundary Cases)**.** *The call price satisfies:*

$$c(t, P_t; \tau, 0) = e^{-r(\tau-t)} P_t, \tag{20}$$

$$c(t, P_t; \tau, 1) = 0. \tag{21}$$

Proof. For $K = 0$: $(P_\tau - 0)^+ = P_\tau$, and $\mathbb{E}^{\mathbb{Q}}[P_\tau \mid P_t] = P_t$ by the martingale property. For $K = 1$: $(P_\tau - 1)^+ = 0$ since $P_\tau \le 1$. □

4.3 European Probability Put Pricing

Theorem 4.3 (European Probability Put Price)**.** *The time-t price of a European probability put is*

$$\mathfrak{p}(t, P_t; \tau, K) = e^{-r(\tau-t)} \int_0^K (K - y)\, p(t, P_t; \tau, y)\, \mathrm{d}y. \tag{22}$$

Corollary 4.4 (Boundary Cases)**.**

$$\mathfrak{p}(t, P_t; \tau, 0) = 0, \tag{23}$$

$$\mathfrak{p}(t, P_t; \tau, 1) = e^{-r(\tau-t)}(1 - P_t). \tag{24}$$

4.4 Put-Call Parity

Theorem 4.5 (Put-Call Parity for Probability Options)**.** *For any strike $K \in [0,1]$ and maturity τ,*

$$c(t, P_t; \tau, K) - \mathfrak{p}(t, P_t; \tau, K) = P_t - Ke^{-r(\tau-t)}. \tag{25}$$

Proof. Using $(P_\tau - K)^+ - (K - P_\tau)^+ = P_\tau - K$ and taking expectations:

$$\begin{aligned} c - \mathfrak{p} &= e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[(P_\tau - K)^+ - (K - P_\tau)^+ \mid P_t] \\ &= e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[P_\tau - K \mid P_t] \\ &= e^{-r(\tau-t)}(P_t - K) = P_t - Ke^{-r(\tau-t)}, \end{aligned}$$

where we used $\mathbb{E}^{\mathbb{Q}}[P_\tau \mid P_t] = P_t$. □

Remark 4.6 (Comparison to Black–Scholes Parity)**.** The standard Black–Scholes put-call parity is $c - \mathfrak{p} = S - Ke^{-r\tau}$. Our formula (25) has the identical structure, with P_t replacing S. The key difference is that $P_t \in [0,1]$, constraining the range of both the underlying and the strike.

4.5 Digital Option Prices

Proposition 4.7 (Digital Option Prices)**.** *The cash-or-nothing call price is*

$$d^{\mathrm{CoN}}(t; \tau, K) = e^{-r(\tau-t)}\mathbb{Q}(P_\tau \geq K \mid P_t) = e^{-r(\tau-t)} \int_K^1 p(t, P_t; \tau, y)\, \mathrm{d}y. \tag{26}$$

The asset-or-nothing call price is

$$d^{\mathrm{AoN}}(t; \tau, K) = e^{-r(\tau-t)} \int_K^1 y\, p(t, P_t; \tau, y)\, \mathrm{d}y. \tag{27}$$

The vanilla call can be decomposed as $c = d^{\mathrm{AoN}} - K \cdot d^{\mathrm{CoN}}$.

4.6 Confidence Digital Prices

Proposition 4.8 (Confidence Digital Prices)**.** *Let $P_{T^-} := \lim_{t \to T^-} P_t$. The upside confidence digital price is*

$$d^{\mathrm{up}}(t; K) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\{E\}}\mathbf{1}_{\{P_{T^-} \geq K\}} \mid \mathcal{F}_t]. \tag{28}$$

Under the model where $\mathbb{Q}(E \mid P_{T^-}) = P_{T^-}$ (the pre-terminal probability is calibrated), this becomes

$$d^{\mathrm{up}}(t; K) = e^{-r(T-t)} \int_K^1 y\, p(t, P_t; T^-, y)\, \mathrm{d}y. \tag{29}$$

Similarly,

$$d^{\mathrm{down}}(t; K) = e^{-r(T-t)} \int_0^K (1-y)\, p(t, P_t; T^-, y)\, \mathrm{d}y. \tag{30}$$

4.7 Belief Variance Swap Strike

Theorem 4.9 (Fair Variance Swap Strike)**.** *Under the martingale Jacobi model, the fair strike for a belief variance swap starting at t_0 with maturity τ is*

$$K^*_{\mathrm{var}} = \mathbb{E}^{\mathbb{Q}}\left[\int_{t_0}^{\tau} \sigma^2 P_s(1 - P_s)\, \mathrm{d}s \,\middle|\, P_{t_0}\right]. \tag{31}$$

Proof. In the continuous limit, $\mathrm{RV}^{(P)} \to [P]$., and the fair strike is the $\mathbb{Q}$-expectation of the realized variance at initiation. □

Proposition 4.10 (Variance Swap Strike Formula)**.** *For the driftless Jacobi process, the fair variance strike is*

$$K^*_{\text{var}} = \sigma^2 \int_{t_0}^{\tau} \mathbb{E}^{\mathbb{Q}}[P_s(1-P_s) \mid P_{t_0}]\,\mathrm{d}s. \tag{32}$$

Since P_s is a martingale, $\mathbb{E}^{\mathbb{Q}}[P_s \mid P_{t_0}] = P_{t_0}$. Let $M_s := \mathbb{E}^{\mathbb{Q}}[P_s^2 \mid P_{t_0}]$. Then

$$K^*_{\text{var}} = \sigma^2 \int_{t_0}^{\tau} (P_{t_0} - M_s)\,\mathrm{d}s. \tag{33}$$

The second moment M_s satisfies the ODE

$$\frac{\mathrm{d}M_s}{\mathrm{d}s} = \sigma^2(P_{t_0} - M_s), \quad M_{t_0} = P_{t_0}^2, \tag{34}$$

which has the closed-form solution

$$M_s = P_{t_0} - P_{t_0}(1-P_{t_0})e^{-\sigma^2(s-t_0)}. \tag{35}$$

Substituting into (33) *and integrating yields*

$$K^*_{\text{var}} = P_{t_0}(1-P_{t_0})\left(1 - e^{-\sigma^2(\tau-t_0)}\right). \tag{36}$$

Proof. Apply Itô's lemma to $f(P) = P^2$ under the dynamics (6):

$$\mathrm{d}(P_s^2) = 2P_s\,\mathrm{d}P_s + \sigma^2 P_s(1-P_s)\,\mathrm{d}s = 2\sigma P_s\sqrt{P_s(1-P_s)}\,\mathrm{d}W_s^{\mathbb{Q}} + \sigma^2 P_s(1-P_s)\,\mathrm{d}s.$$

Taking $\mathbb{Q}$-expectations and using the martingale property of the stochastic integral:

$$\frac{\mathrm{d}M_s}{\mathrm{d}s} = \sigma^2\mathbb{E}^{\mathbb{Q}}[P_s(1-P_s) \mid P_{t_0}] = \sigma^2(P_{t_0} - M_s).$$

This linear ODE with initial condition $M_{t_0} = P_{t_0}^2$ has solution (35). The variance swap strike then follows by direct integration. □

Remark 4.11 (Interpretation of the Variance Swap Strike)**.** Formula (36) has an intuitive structure. The leading factor $P_{t_0}(1-P_{t_0})$ is the instantaneous variance rate (maximized when $P_{t_0} = 1/2$, i.e., maximum uncertainty). The exponential term captures the decay of expected future variance: as $\tau - t_0$ grows, $M_s \to P_{t_0}$ (the second moment approaches the squared first moment, meaning $\mathrm{Var}(P_s) \to 0$ as information resolves), and the fair strike asymptotes to $P_{t_0}(1-P_{t_0})$.

4.8 Model-Free Replication of Belief Variance Swaps

In equity markets, the celebrated result of Neuberger (1994) shows that variance swaps can be replicated by a static position in the log contract. We derive the bounded-domain analog for belief variance swaps.

Theorem 4.12 (Variance Swap Replication)**.** *Define the* bounded log payoff

$$L(P) := -\log P - \log(1-P). \tag{37}$$

Then for any continuous semimartingale $P_t \in (0,1)$,

$$L(P_\tau) - L(P_{t_0}) = -\int_{t_0}^{\tau} \frac{1-2P_s}{P_s(1-P_s)}\,\mathrm{d}P_s + \int_{t_0}^{\tau} \frac{1}{P_s(1-P_s)}\,\mathrm{d}[P]_s. \tag{38}$$

Under the Jacobi dynamics, $\mathrm{d}[P]_s = \sigma^2 P_s(1-P_s)\,\mathrm{d}s$*, so*

$$\int_{t_0}^{\tau} \sigma^2 \,\mathrm{d}s = L(P_\tau) - L(P_{t_0}) + \int_{t_0}^{\tau} \frac{1-2P_s}{P_s(1-P_s)}\,\mathrm{d}P_s. \tag{39}$$

More generally, without assuming the Jacobi model, the realized quadratic variation satisfies

$$[P]_{t_0,\tau} = \int_{t_0}^{\tau} P_s(1-P_s)\left(L(P_\tau) - L(P_{t_0}) + \int_{t_0}^{\tau} \frac{1-2P_u}{P_u(1-P_u)}\,\mathrm{d}P_u\right)\frac{\mathrm{d}[P]_s}{\sigma^2\,\mathrm{d}s}. \tag{40}$$

Proof. Apply Itô's lemma to $L(P) = -\log P - \log(1-P)$:

$$\begin{aligned}
\mathrm{d}L(P_t) &= L'(P_t)\,\mathrm{d}P_t + \tfrac{1}{2}L''(P_t)\,\mathrm{d}[P]_t,\\
L'(P) &= -\frac{1}{P} + \frac{1}{1-P} = \frac{2P-1}{P(1-P)},\\
L''(P) &= \frac{1}{P^2} + \frac{1}{(1-P)^2} = \frac{1}{P^2(1-P)^2}.
\end{aligned}$$

Since $\mathrm{d}[P]_t = \sigma^2 P_t(1-P_t)\,\mathrm{d}t$ under the Jacobi model:

$$\tfrac{1}{2}L''(P_t)\,\mathrm{d}[P]_t = \frac{\sigma^2}{2P_t(1-P_t)}\,\mathrm{d}t.$$

Rearranging the Itô expansion gives (38). □

Remark 4.13 (Static Replication via Option Strip)**.** The bounded log payoff $L(P) = -\log P - \log(1-P)$ can be statically replicated by a strip of European probability options. Using the Carr–Madan decomposition on $[0,1]$, for any twice-differentiable f and reference point P^*:

$$f(P_\tau) = f(P^*) + f'(P^*)(P_\tau - P^*) + \int_0^{P^*} f''(K)(K-P_\tau)^+\,\mathrm{d}K + \int_{P^*}^{1} f''(K)(P_\tau - K)^+\,\mathrm{d}K.$$

Applying this to $f = L$ with $L''(K) = K^{-2} + (1-K)^{-2}$, the bounded log payoff is replicated by the event future (for the linear term) plus a continuum of puts (for $K < P^*$) and calls (for $K > P^*$), weighted by $L''(K)$. This yields model-free pricing of belief variance swaps through observed option prices, without requiring the Jacobi model to be correctly specified.

5 Greeks and Hedging

5.1 Definition of Greeks

For a smooth pricing function $V = V(t, P; \sigma, r, K, \tau)$, we define the following sensitivities:

Definition 5.1 (Probability Greeks)**.**

$$\Delta_P := \frac{\partial V}{\partial P}, \qquad \text{(delta: sensitivity to underlying probability)} \tag{41}$$

$$\Gamma_P := \frac{\partial^2 V}{\partial P^2}, \qquad \text{(gamma: convexity in probability)} \tag{42}$$

$$\mathcal{V}_P := \frac{\partial V}{\partial \sigma}, \qquad \text{(vega: sensitivity to belief volatility)} \tag{43}$$

$$\Theta := \frac{\partial V}{\partial t}, \qquad \text{(theta: time decay)} \tag{44}$$

$$\rho := \frac{\partial V}{\partial r}. \qquad \text{(rho: interest rate sensitivity)} \tag{45}$$

5.2 Theta–Gamma Relationship

Proposition 5.2 (Theta–Gamma–Value Identity)**.** *For any European claim priced under the Jacobi model, the Greeks satisfy*

$$\Theta + \tfrac{1}{2}\sigma^2 P(1-P)\,\Gamma_P = rV. \tag{46}$$

Equivalently, time decay equals the risk-free carry minus the gamma contribution:

$$\Theta = rV - \tfrac{1}{2}\sigma^2 P(1-P)\,\Gamma_P. \tag{47}$$

Proof. This is an immediate rearrangement of the pricing PDE (52). □

Remark 5.3. Identity (46) is the bounded-domain analog of the classical relationship $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma = rV$ from Black–Scholes. It governs the "theta bleed" of option positions: a long gamma position ($\Gamma_P > 0$) earns from realized moves in P_t but pays theta; the PDE ensures these exactly offset under continuous hedging. In the bounded setting, note that the gamma contribution $\frac{1}{2}\sigma^2 P(1-P)\Gamma_P$ is maximized near $P = 1/2$ and vanishes at the boundaries, reflecting the reduced optionality when the probability is near certainty.

5.3 Delta for European Calls

Proposition 5.4 (Call Delta)**.** *The delta of a European probability call is*

$$\Delta_c = e^{-r(\tau-t)} \int_K^1 \frac{\partial p(t, P_t; \tau, y)}{\partial P_t}(y-K)\,\mathrm{d}y. \tag{48}$$

At the boundaries:

- *As $P_t \to 0$: $\Delta_c \to 0$ (call is worthless).*
- *As $P_t \to 1$: $\Delta_c \to e^{-r(\tau-t)}$ (call is deeply in the money).*

Proof. Differentiate (19) under the integral sign. The boundary behavior follows from $P_\tau \to 0$ or $P_\tau \to 1$ as P_t approaches the boundaries. □

Corollary 5.5 (Put Delta)**.** *By put-call parity* (25)*,*

$$\Delta_{\mathfrak{p}} = \Delta_c - 1. \tag{49}$$

5.4 Gamma

Proposition 5.6 (Call Gamma)**.** *The gamma of a European probability call is*

$$\Gamma_c = e^{-r(\tau-t)} \int_K^1 \frac{\partial^2 p(t, P_t; \tau, y)}{\partial P_t^2}(y-K)\,\mathrm{d}y. \tag{50}$$

Since c is convex in P_t, we have $\Gamma_c \geq 0$. Maximum gamma occurs near $P_t = K$.

5.5 Vega

Proposition 5.7 (Call Vega)**.** *The vega of a European probability call is*

$$\mathcal{V}_c = e^{-r(\tau-t)} \int_K^1 \frac{\partial p(t, P_t; \tau, y)}{\partial \sigma}(y-K)\,\mathrm{d}y > 0. \tag{51}$$

Higher belief volatility increases option value (for both calls and puts).

5.6 Delta Hedging Strategy

Theorem 5.8 (Delta-Hedging Portfolio)**.** *Consider a short position in a European probability call. The replicating portfolio holds:*

- Δ_c *units of the event future,*
- $c - \Delta_c \cdot P_t$ *invested in the money market.*

The portfolio value matches the call value, and under continuous rebalancing, the strategy replicates the call payoff.

Proof. This is the standard Black–Scholes hedging argument adapted to the bounded underlying. Let $\Pi_t = -c + \Delta_c P_t + \xi_t B_t$ be the hedged portfolio with ξ_t chosen so $\Pi_t = 0$. Applying Itô's lemma to $c(t, P_t)$ and P_t:

$$\mathrm{d}c = \Theta\,\mathrm{d}t + \Delta_c\,\mathrm{d}P_t + \tfrac{1}{2}\Gamma_c\sigma^2 P_t(1 - P_t)\,\mathrm{d}t.$$

The portfolio P&L is

$$\mathrm{d}\Pi = -\mathrm{d}c + \Delta_c\,\mathrm{d}P_t + r\xi_t B_t\,\mathrm{d}t.$$

Substituting and using $\Pi_t = 0$:

$$\mathrm{d}\Pi = \big(-\Theta - \tfrac{1}{2}\Gamma_c\sigma^2 P_t(1 - P_t) + r(c - \Delta_c P_t)\big)\,\mathrm{d}t.$$

The pricing PDE (52) ensures $\mathrm{d}\Pi = 0$, confirming the hedge is self-financing and replicating. □

5.7 The Pricing PDE

Theorem 5.9 (Black–Scholes PDE for Probability Options)**.** *Under the martingale Jacobi model, the price $V(t, P)$ of any European claim with payoff $H(P_\tau)$ satisfies*

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 P(1 - P)\frac{\partial^2 V}{\partial P^2} - rV = 0, \quad (t, P) \in [0, \tau) \times (0, 1), \tag{52}$$

with terminal condition $V(\tau, P) = H(P)$ and boundary conditions $V(t, 0) = e^{-r(\tau - t)}H(0)$, $V(t, 1) = e^{-r(\tau - t)}H(1)$.

Proof. Apply the Feynman–Kac theorem to the risk-neutral pricing formula (18). The driftless diffusion coefficient $\frac{1}{2}\sigma^2 P(1 - P)$ enters the second-order term. □

Remark 5.10. The PDE (52) differs from the standard Black–Scholes PDE ($\frac{1}{2}\sigma^2 S^2 V_{SS}$) in the diffusion coefficient. The $P(1 - P)$ term reflects the bounded nature of the underlying.

6 Market Completeness

We now establish that the OIEC market is complete: any bounded contingent claim can be replicated.

6.1 Continuous-Time Completeness

Theorem 6.1 (Market Completeness)**.** *Consider a market with:*

(i) A money market account B_t,

(ii) An event future with price P_t following (6)*.*

This market is complete: any $\mathcal{F}_\tau$-measurable, bounded contingent claim H can be replicated by a self-financing trading strategy in the event future and money market.

Proof. The martingale Jacobi process on $(0,1)$ generates a filtration $(\mathcal{F}_t^P)$ that equals $(\mathcal{F}_t^W)$, the filtration generated by the driving Brownian motion. This is because the diffusion coefficient $\sigma\sqrt{P_t(1-P_t)} > 0$ for $P_t \in (0,1)$, so the Brownian motion can be recovered from P_t by $\mathrm{d}W_t^{\mathbb{Q}} = [\sigma\sqrt{P_t(1-P_t)}]^{-1}\mathrm{d}P_t$. By the martingale representation theorem, any $\mathcal{F}_\tau$-measurable square-integrable claim H can be written as

$$e^{-r\tau}H = V_0 + \int_0^\tau \phi_s \,\mathrm{d}\tilde{P}_s$$

for some predictable process ϕ. This integral representation defines the replicating strategy: hold ϕ_t units of the event future and invest the remainder in the money market. □

Corollary 6.2 (Uniqueness of the Risk-Neutral Measure)**.** *Under the Jacobi diffusion model, the equivalent martingale measure $\mathbb{Q}$ is unique. Consequently, every contingent claim has a unique arbitrage-free price.*

Proof. Market completeness is equivalent to uniqueness of the EMM by the second fundamental theorem of asset pricing (Harrison and Pliska, 1981). Since any claim can be replicated (Theorem 6.1), the pricing functional is unique, implying $\mathbb{Q}$ is the only EMM. □

6.2 Discrete-Time Completeness

For numerical implementation, consider a discrete approximation.

Definition 6.3 (Recombining Tree)**.** Fix integers $N, M \geq 1$. Define:

- Time steps: $\Delta t = \tau/N$, with $t_n = n\Delta t$ for $n = 0, \ldots, N$.
- State space: $\mathcal{G} = \{k/M : k = 0, 1, \ldots, M\} \subset [0,1]$.
- Transition probabilities: $q(p, p')$ for $p, p' \in \mathcal{G}$ chosen to match the Jacobi diffusion moments.

Theorem 6.4 (Discrete Market Completeness)**.** *In the recombining tree model with $M+1$ states and trading in the event future and money market, the market is complete if and only if the number of traded assets equals the number of successor states at each node.*

Proof. At each node $p \in \mathcal{G}$ at time t_n, the event future can transition to at most 3 successor states (up, middle, down in a trinomial tree). With 2 assets (event future + money market), we can replicate any claim on 2 successor states. For 3 states, we add one European option to span the third dimension. □

Corollary 6.5. *In a binomial approximation to the Jacobi process (2 successor states per node), the market consisting of the event future and money market alone is complete.*

7 Extensions

7.1 American Probability Options

American options allow early exercise, which may be valuable given the bounded nature of the underlying.

Definition 7.1 (American Probability Put)**.** The holder of an American probability put with strike K can exercise at any stopping time $\tau^* \leq \tau$, receiving payoff $(K - P_{\tau^*})^+$.

Theorem 7.2 (American Put Value and Free-Boundary Problem)**.** *The American probability put price is*

$$\mathfrak{p}^{\mathrm{Am}}(t, P_t; \tau, K) = \sup_{\tau^* \in \mathcal{T}_{t,\tau}} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau^* - t)} (K - P_{\tau^*})^+ \mid P_t \right], \tag{53}$$

where $\mathcal{T}_{t,\tau}$ is the set of stopping times valued in $[t, \tau]$. Equivalently, $\mathfrak{p}^{\mathrm{Am}}$ satisfies the linear complementarity problem

$$\max\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 P(1-P) \frac{\partial^2 V}{\partial P^2} - rV,\ (K - P) - V \right) = 0, \tag{54}$$

on $(t, P) \in [0, \tau) \times (0, 1)$, with terminal condition $V(\tau, P) = (K - P)^+$.

Proof. The optimal stopping representation (53) is standard. The complementarity formulation follows from the general theory of optimal stopping for diffusions: in the continuation region $\{(t, P) : V(t, P) > (K - P)^+\}$, the pricing PDE holds; in the exercise region $\{(t, P) : V(t, P) = (K - P)^+\}$, the PDE inequality holds with the payoff constraint binding. □

Proposition 7.3 (Early Exercise Premium)**.** *For $r > 0$, early exercise of the American probability put may be optimal when P_t is sufficiently close to 0. The critical boundary $P^*(t)$ separating the exercise and continuation regions satisfies*

$$\mathfrak{p}^{\mathrm{Am}}(t, P^*(t); \tau, K) = K - P^*(t). \tag{55}$$

At the boundary, the smooth-pasting condition holds: $\partial \mathfrak{p}^{\mathrm{Am}} / \partial P \big|_{P = P^*(t)} = -1$.

Proof. When $P_t \approx 0$, the volatility $\sigma\sqrt{P_t(1 - P_t)} \approx \sigma\sqrt{P_t} \to 0$, so the process moves slowly and the option is deep in the money. Continuing to hold the put earns approximately $rK\,\mathrm{d}t$ in time value of money but risks P_t increasing. If the carrying cost exceeds the expected option appreciation, immediate exercise is optimal. The smooth-pasting condition is standard for American options on diffusions and ensures optimality of the boundary. □

7.2 Barrier Options

Definition 7.4 (Knock-Out Probability Call)**.** An *up-and-out probability call* with barrier $H \in (K, 1)$ pays $(P_\tau - K)^+ \mathbf{1}_{\{\max_{t \le s \le \tau} P_s < H\}}$.

Barrier options on bounded processes have well-defined prices since the underlying cannot exceed 1.

7.3 Spread Options on Correlated Events

Consider two events E_1, E_2 with event futures $P_t^{(1)}, P_t^{(2)}$.

Definition 7.5 (Bivariate Jacobi Dynamics)**.** Let $(P_t^{(1)}, P_t^{(2)})$ follow correlated martingale Jacobi processes:

$$\mathrm{d}P_t^{(1)} = \sigma_1 \sqrt{P_t^{(1)}(1 - P_t^{(1)})}\, \mathrm{d}W_t^{(1)}, \tag{56}$$

$$\mathrm{d}P_t^{(2)} = \sigma_2 \sqrt{P_t^{(2)}(1 - P_t^{(2)})}\, \mathrm{d}W_t^{(2)}, \tag{57}$$

where $\mathrm{d}W_t^{(1)} \mathrm{d}W_t^{(2)} = \rho\,\mathrm{d}t$ with correlation $\rho \in [-1, 1]$.

Definition 7.6 (Probability Spread Option)**.** A spread call with maturity τ pays $(P_\tau^{(1)} - P_\tau^{(2)} - K)^+$.

Proposition 7.7 (Spread Option Pricing)**.** *The spread call price is*

$$c^{\text{spread}}(t) = e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}\left[(P_\tau^{(1)} - P_\tau^{(2)} - K)^+ \,\middle|\, \mathcal{F}_t\right], \tag{58}$$

which requires evaluation of the bivariate transition density $p^{(12)}(t, x_1, x_2; \tau, y_1, y_2)$ of the correlated Jacobi system. No closed-form solution exists in general; pricing proceeds via Monte Carlo simulation of the system (56)–(57) *using the Cholesky decomposition $W_t^{(2)} = \rho W_t^{(1)} + \sqrt{1-\rho^2}\,\widetilde{W}_t$ for independent $\widetilde{W}_t$.*

Remark 7.8 (Correlation Calibration)**.** The correlation parameter ρ can be estimated from the historical covariance of the two event futures. Define the realized covariance estimator

$$\hat{\rho} = \frac{\sum_{i=1}^{n-1} \Delta P_{t_i}^{(1)} \Delta P_{t_i}^{(2)}}{\sqrt{\sum_{i=1}^{n-1} (\Delta P_{t_i}^{(1)})^2 \sum_{i=1}^{n-1} (\Delta P_{t_i}^{(2)})^2}},$$

where $\Delta P_{t_i}^{(j)} = P_{t_{i+1}}^{(j)} - P_{t_i}^{(j)}$. Alternatively, if spread options trade, ρ can be implied from observed prices. The multivariate Jacobi framework of Gouriéroux and Jasiak (2006) provides further structure for correlation modeling in this setting.

7.4 Conditional Event Options

Definition 7.9 (Conditional Option)**.** Let E_1, E_2 be two events with $T_1 < T_2$. A conditional call pays

$$(P_{T_1}^{(2)} - K)^+ \cdot \mathbf{1}_{\{E_1\}},$$

i.e., the payoff depends on the probability of E_2 at the resolution time of E_1, conditional on E_1 occurring.

8 Numerical Methods

8.1 Trinomial Tree for the Jacobi Process

We construct a recombining trinomial tree that matches the first two moments of the Jacobi diffusion at each node.

Algorithm 1 Trinomial Tree for Jacobi Process

Require: Initial probability P_0, volatility σ, maturity τ, steps N, states M

1: $\Delta t \leftarrow \tau/N$
2: $\Delta P \leftarrow 1/M$
3: **for** each node (n, k) with $P = k\Delta P$, $n < N$ **do**
4: Compute local variance $v \leftarrow \sigma^2 P(1-P)\Delta t$
5: Solve for (q_u, q_m, q_d) matching:

$$\begin{aligned} q_u + q_m + q_d &= 1 \\ q_u \Delta P - q_d \Delta P &= 0 \quad \text{(martingale)} \\ q_u (\Delta P)^2 + q_d (\Delta P)^2 &= v \quad \text{(variance)} \end{aligned}$$

6: Store transitions: up to $\min(k+1, M)$, middle at k, down to $\max(k-1, 0)$
7: **end for**

Proposition 8.1 (Well-Posedness)**.** *The trinomial system has a unique solution with* $q_u, q_m, q_d \in (0,1)$ *provided*

$$\sigma^2 \Delta t < \frac{(\Delta P)^2}{P(1-P)}. \tag{59}$$

This is always satisfiable for sufficiently small Δt*.*

8.2 Backward Induction for Option Pricing

Algorithm 2 European Option Pricing via Backward Induction

Require: Tree from Algorithm 1, strike K, discount factor $\delta = e^{-r\Delta t}$

```
Initialize: V_N(k) ← (kΔP − K)^+ for all k = 0, ..., M
for n = N − 1 down to 0 do
    for k = 0 to M do
        V_n(k) ← δ (q_u V_{n+1}(k + 1) + q_m V_{n+1}(k) + q_d V_{n+1}(k − 1))
    end for
end for
return V_0(k_0) where k_0 = ⌊P_0/ΔP⌉
```

8.3 Finite Difference Methods

The PDE (52) can be solved via finite differences.

Definition 8.2 (Crank–Nicolson Scheme)**.** Define grid $P_j = j\Delta P$, $t_n = n\Delta t$, and let $V_j^n \approx V(t_n, P_j)$. The Crank–Nicolson scheme is

$$\frac{V_j^{n+1} - V_j^n}{\Delta t} = \frac{1}{2}\left(LV_j^{n+1} + LV_j^n\right), \tag{60}$$

where L is the spatial operator

$$LV_j = \frac{\sigma^2}{2} P_j(1-P_j)\frac{V_{j+1} - 2V_j + V_{j-1}}{(\Delta P)^2} - rV_j. \tag{61}$$

Proposition 8.3 (Stability)**.** *The Crank–Nicolson scheme is unconditionally stable and second-order accurate in both* Δt *and* ΔP*.*

8.4 Monte Carlo Simulation

For path-dependent options (barriers, variance swaps), Monte Carlo methods are efficient.

Algorithm 3 Monte Carlo for Jacobi Process

Require: P_0, σ, τ, N steps, M paths

```
Δt ← τ/N
for m = 1 to M do
    P ← P_0
    for n = 1 to N do
        Z ← N(0, 1) sample
        P ← P + σ √(P(1 − P)Δt) · Z
        P ← max(0, min(1, P))                    ▷ Boundary truncation
    end for
    Store P_τ^(m) ← P
end for
return Sample mean of payoffs
```

Remark 8.4 (Euler–Maruyama Bias)**.** The truncation at boundaries introduces bias. More accurate schemes include the Milstein method or exact simulation of the Beta bridge. For the Jacobi martingale, the Milstein correction is

$$P_{n+1} = P_n + \sigma\sqrt{P_n(1-P_n)\Delta t}\, Z + \frac{\sigma^2}{2}(1-2P_n)\Delta t\,(Z^2-1),$$

which improves strong convergence from order 1/2 to order 1.

9 Calibration

9.1 Observable Market Data

Calibration of the OIEC model requires:

(i) **Event future prices:** The current level P_t from the prediction market.

(ii) **Historical prices:** A time series $(P_{t_1}, \ldots, P_{t_n})$ for realized volatility estimation.

(iii) **Option prices:** If probability options trade, their prices across strikes and maturities.

9.2 Volatility Estimation from Historical Data

Definition 9.1 (Realized Belief Volatility)**.** Given observations $P_{t_1}, \ldots, P_{t_n}$ with $\Delta t = t_{i+1} - t_i$, the realized belief volatility estimator is

$$\hat{\sigma}^2 = \frac{1}{(n-1)\Delta t}\sum_{i=1}^{n-1}\frac{(P_{t_{i+1}} - P_{t_i})^2}{P_{t_i}(1-P_{t_i})}. \tag{62}$$

Proposition 9.2 (Consistency)**.** *Under the Jacobi model, $\hat{\sigma}^2 \to \sigma^2$ in probability as $n \to \infty$ with $\Delta t \to 0$ and $(n-1)\Delta t \to \infty$.*

Proof sketch. Under (6), the increments satisfy $P_{t_{i+1}} - P_{t_i} = \sigma \int_{t_i}^{t_{i+1}} \sqrt{P_s(1-P_s)}\,\mathrm{d}W_s^{\mathbb{Q}}$. By Itô isometry,

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{(P_{t_{i+1}} - P_{t_i})^2}{P_{t_i}(1-P_{t_i})}\right] = \frac{\sigma^2}{P_{t_i}(1-P_{t_i})}\int_{t_i}^{t_{i+1}} \mathbb{E}^{\mathbb{Q}}[P_s(1-P_s) \mid P_{t_i}]\,\mathrm{d}s \xrightarrow{\Delta t \to 0} \sigma^2 \Delta t.$$

The ratio converges because $\mathbb{E}^{\mathbb{Q}}[P_s(1-P_s) \mid P_{t_i}] \to P_{t_i}(1-P_{t_i})$ as $s \to t_i$. Summing and dividing by $(n-1)\Delta t$, the law of large numbers for martingale differences yields $\hat{\sigma}^2 \to \sigma^2$ in probability. □

Remark 9.3 (Practical Considerations for Volatility Estimation)**.** The estimator (62) has a singularity when P_{t_i} is near 0 or 1, since the denominator $P_{t_i}(1-P_{t_i})$ approaches zero. In practice, observations with $P_{t_i} < \varepsilon$ or $P_{t_i} > 1-\varepsilon$ (for some small $\varepsilon > 0$, e.g., $\varepsilon = 0.02$) should be excluded from the estimation, or the estimator should be winsorized. Additionally, prediction market prices exhibit features not captured by the constant-σ model, including jumps on news events (e.g., debate outcomes, economic data releases), intraday liquidity variation, and bid-ask bounce effects. Robustness can be improved by using a realized variance estimator based on higher-frequency data with noise corrections.

9.3 Implied Volatility from Option Prices

If European probability options trade, implied volatility $\sigma^{\text{imp}}(K, \tau)$ is defined as the value of σ that equates the model price (19) to the observed market price.

Definition 9.4 (Implied Belief Volatility)**.** Given observed call price c^{mkt} for strike K and maturity τ, the implied belief volatility σ^{imp} satisfies

$$c(t, P_t; \tau, K; \sigma^{\text{imp}}) = c^{\text{mkt}}. \tag{63}$$

The implied volatility is found via root-finding (e.g., Newton–Raphson) using vega as the derivative.

9.4 Volatility Surface Calibration

With multiple option prices across strikes and maturities, we fit a local volatility function $\sigma(P, t)$ by minimizing pricing errors.

Definition 9.5 (Calibration Objective)**.**

$$\min_{\sigma(\cdot,\cdot)} \sum_i w_i \left(c^{\text{model}}(K_i, \tau_i; \sigma) - c_i^{\text{mkt}} \right)^2 + \lambda \|\sigma\|_{\text{reg}}^2, \tag{64}$$

where w_i are weights (e.g., inverse vega), and $\lambda\|\sigma\|_{\text{reg}}^2$ is a regularization term penalizing irregular volatility surfaces.

9.5 Calibration to Event Characteristics

For events with known structural features, volatility can be parameterized accordingly.

Example 9.6 (Election Volatility)**.** For political elections, volatility typically peaks on election day and is higher during debate periods. A seasonal model is

$$\sigma(t) = \sigma_0 + \sigma_1 \mathbf{1}_{\{t \in \text{debate window}\}} + \sigma_2 \cdot \max\left(0, 1 - \frac{T - t}{\delta}\right), \tag{65}$$

where δ is a terminal ramp-up period.

10 Numerical Example

We illustrate the framework with a concrete calibration and pricing exercise based on stylized prediction market data.

10.1 Setup

Consider a hypothetical event future on "Fed raises rates at the March 2027 meeting." The current date is $t = 0$, event maturity is $T = 0.25$ years (3 months), and the current event future price is $P_0 = 0.60$. The risk-free rate is $r = 5\%$. We consider European probability options with maturity $\tau = 0.20$ years (approximately 10 weeks before the event, ensuring $\tau < T$).

10.2 Calibration

From a hypothetical 60-day history of hourly event future prices (approximately $n = 480$ observations at $\Delta t = 1/6720$ years), suppose the realized belief volatility estimator (62) yields $\hat{\sigma} = 1.50$. This implies an instantaneous standard deviation of $\sigma\sqrt{P_0(1 - P_0)} = 1.50 \times \sqrt{0.60 \times 0.40} \approx 0.735$ in annualized probability units.

10.3 Option Prices Across Strikes

Using the Crank–Nicolson scheme (Section 8.2) with $M = 200$ spatial nodes and $N = 500$ time steps, we compute European probability call and put prices across strikes $K \in \{0.10, 0.20, \ldots, 0.90\}$. Table 1 reports the results.

Table 1: European probability option prices. Parameters: $P_0 = 0.60$, $\sigma = 1.50$, $\tau = 0.20$, $r = 5\%$.

Strike K	Call c	Put $\mathfrak{p}$	Parity check $c - \mathfrak{p}$	$P_0 - Ke^{-r\tau}$
0.10	0.5010	0.0011	0.4999	0.5010
0.20	0.4073	0.0074	0.3999	0.4010
0.30	0.3196	0.0197	0.2999	0.3010
0.40	0.2402	0.0403	0.1999	0.2010
0.50	0.1718	0.0719	0.0999	0.1010
0.60	0.1159	0.1160	−0.0001	0.0010
0.70	0.0726	0.1727	−0.1001	−0.0990
0.80	0.0402	0.2403	−0.2001	−0.1990
0.90	0.0166	0.3167	−0.3001	−0.2990

The parity check column confirms put-call parity (25) holds to numerical precision.

10.4 Greeks Profile

Table 2 reports the Greeks for the at-the-money call ($K = 0.60$) across different underlying probability levels.

Table 2: Greeks for European probability call at $K = 0.60$, $\sigma = 1.50$, $\tau = 0.20$, $r = 5\%$.

P_t	Δ_c	Γ_c	$\mathcal{V}_c$	Θ_c
0.20	0.1512	0.6421	0.0320	−0.0189
0.40	0.3587	0.9048	0.0765	−0.0591
0.50	0.4694	0.9370	0.0903	−0.0739
0.60	0.5728	0.9048	0.0936	−0.0799
0.80	0.7988	0.6421	0.0632	−0.0565

Key observations: gamma is maximized near $P_t = K = 0.60$ (at-the-money), vega peaks slightly below ATM due to the asymmetry of $P(1 - P)$, and theta is uniformly negative (the option loses value as time passes), consistent with the identity (46).

10.5 Variance Swap Strike

The fair belief variance swap strike from (36) is

$$K^*_{\text{var}} = 0.60 \times 0.40 \times \left(1 - e^{-1.50^2 \times 0.20}\right) = 0.24 \times (1 - e^{-0.45}) \approx 0.24 \times 0.3624 = 0.0870.$$

This represents the fair price (in probability-squared units) of realized belief variance over the option horizon. For comparison, the instantaneous variance rate is $\sigma^2 P_0(1 - P_0) = 2.25 \times 0.24 = 0.54$ per year, and the naive annualized estimate $0.54 \times 0.20 = 0.108$ overstates the fair strike because it ignores the decay of expected variance captured by the exponential term.

11 Product Specification

This section provides a prototype term sheet for the simplest OIEC: a European probability call on an event future. The specification follows the format used by CFTC-regulated derivatives exchanges.

11.1 Contract Terms

Table 3: Prototype OIEC product specification: European probability call.

Field	**Specification**
Underlying	Event future on E, exchange-listed
Contract type	European-style probability call
Contract size	1,000 contracts (notional = 1,000 × \$1 = \$1,000)
Tick size	\$0.001 (0.1 cent per contract, \$1.00 per lot)
Strike increments	0.05 (i.e., $K \in \{0.05, 0.10, \ldots, 0.95\}$)
Listed maturities	Weekly and monthly, up to 3 months out
Last trading day	2 business days before event maturity T
Exercise	Cash-settled, automatic at expiry
Settlement price	Closing price P_τ of underlying event future
Payoff per contract	$\max(P_\tau - K, 0) \times \$1{,}000$
Trading hours	Same as underlying event future
Position limits	50,000 lots per account (subject to exchange review)

11.2 Design Rationale

The contract multiplier of 1,000 is chosen to balance accessibility (a contract at $P = 0.50$, $K = 0.50$ costs roughly \$100–\$150 depending on vol and time) against the need for meaningful notional for institutional use. The tick size of \$0.001 gives sufficient price granularity for market makers to quote inside the bid-ask spread. Strike increments of 0.05 provide 19 strikes spanning the full probability range; finer increments (0.01) could be introduced as liquidity develops.

The last trading day is set 2 business days before event maturity to avoid the degenerate pricing regime as P_t pins toward $\{0, 1\}$ (Remark 2.13). This also allows orderly settlement and reduces the risk of manipulation near the terminal date.

11.3 Additional Products

The same specification extends naturally to:

- **Probability puts:** Identical terms, payoff $\max(K - P_\tau, 0) \times \$1{,}000$.
- **Belief variance swaps:** OTC or exchange-listed, payoff $(\mathrm{RV}^{(P)} - K_{\mathrm{var}}) \times$ notional, settled against realized quadratic variation of the underlying event future over the contract period.
- **Straddles/strangles:** Packaged combinations, traded as single orders.

12 Margin Methodology

A clearinghouse listing OIECs needs a margin framework that reflects the bounded nature of the underlying. We propose a methodology based on Greeks exposure, with a structure similar to SPAN but adapted to $[0, 1]$.

12.1 Maximum Loss Bounds

The bounded underlying provides hard limits on potential loss that do not exist for equity options.

Proposition 12.1 (Maximum Loss for Short Positions). *(i) Short call: maximum loss is* $(1-K) \times$ *multiplier, attained when* $P_\tau = 1$.

(ii) Short put: maximum loss is $K \times$ *multiplier, attained when* $P_\tau = 0$.

(iii) Short straddle at K*: maximum loss is* $\max(K, 1-K) \times$ *multiplier.*

All losses are bounded by \$1,000 per lot (the contract multiplier times the maximum price range of 1).

This is a fundamental advantage over equity options, where losses on short calls are theoretically unlimited.

12.2 Initial Margin

We propose initial margin as the maximum of two components:

Definition 12.2 (OIEC Initial Margin). For a single position of n lots:

$$\mathrm{IM} = n \times \text{multiplier} \times \max(\mathrm{IM}_{\text{scan}},\ \mathrm{IM}_{\text{floor}})\,, \tag{66}$$

where:

$$\mathrm{IM}_{\text{scan}} = |\Delta_P| \cdot \Delta P^{\max} + \tfrac{1}{2}|\Gamma_P| \cdot (\Delta P^{\max})^2 + |\mathcal{V}_P| \cdot \Delta\sigma^{\max}, \tag{67}$$

$$\mathrm{IM}_{\text{floor}} = f \cdot \text{max_loss}. \tag{68}$$

Here $\Delta P^{\max}$ is the maximum one-day probability move (a scan range), $\Delta\sigma^{\max}$ is a vol shock, and f is a floor fraction (e.g., $f = 0.10$).

12.3 Scan Range Calibration

The scan range $\Delta P^{\max}$ should be calibrated to historical tail moves. For an event future at P_t with volatility σ, the 1-day 99% VaR under the Jacobi model is approximately

$$\Delta P^{\max}(P_t) = 2.33 \times \sigma\sqrt{P_t(1-P_t)/252}. \tag{69}$$

Note that this is state-dependent: the scan range shrinks near boundaries. At $P_t = 0.50$ with $\sigma = 1.50$, the 1-day 99% scan range is $2.33 \times 1.50 \times \sqrt{0.25/252} \approx 0.110$. At $P_t = 0.90$, it drops to $2.33 \times 1.50 \times \sqrt{0.09/252} \approx 0.066$.

Remark 12.3 (Jump Risk Margin Add-On). The Jacobi model underestimates tail risk on days with scheduled news events (debates, FOMC meetings, data releases). On these days, the exchange should apply a margin multiplier (e.g., 1.5× to 2×) or use a historically calibrated jump-adjusted scan range. This parallels the treatment of earnings announcements in equity option margin.

12.4 Portfolio Margin

For accounts holding multiple OIECs on the same underlying, offsets apply:

- Long call + short put at same strike and maturity: margin equals event future margin (synthetic future).
- Spreads (bull call, bear put): margin equals maximum spread loss.
- Straddles/strangles: margin based on net Greeks.

12.5 Variation Margin

Daily mark-to-market using the pricing PDE (52) or the exchange settlement price. Variation margin is the daily change in position value, collected from losing accounts and paid to gaining accounts.

12.6 Numerical Example: Margin Calculation

Consider a short position of 10 lots in a probability call with $K = 0.50$, $P_t = 0.60$, $\sigma = 1.50$, $\tau = 0.20$, $r = 5\%$.

From Table 2: $\Delta_c = 0.5728$, $\Gamma_c = 0.9048$, $\mathcal{V}_c = 0.0936$.

With $\Delta P^{\max} = 0.110$ and $\Delta\sigma^{\max} = 0.30$:

$$\begin{aligned} \mathrm{IM}_{\mathrm{scan}} &= 0.5728 \times 0.110 + 0.5 \times 0.9048 \times 0.110^2 + 0.0936 \times 0.30 \\ &= 0.0630 + 0.0055 + 0.0281 = 0.0966. \end{aligned}$$

Maximum loss for a short call at $K = 0.50$: $(1 - 0.50) = 0.50$, so $\mathrm{IM}_{\mathrm{floor}} = 0.10 \times 0.50 = 0.05$.

Initial margin per lot: $\$1{,}000 \times \max(0.0966, 0.05) = \$96.60$.

Total for 10 lots: \$966.

13 Market-Making Economics

A critical question for exchange viability: can market makers earn enough to justify quoting these products?

13.1 Revenue Model

Proposition 13.1 (Market Maker P&L: Detailed)**.** *A market maker quoting bid-ask spread s who delta-hedges earns approximately*

$$P\&L \approx s \cdot Q - \frac{1}{2}\int_0^\tau \Gamma \cdot \sigma_{\mathrm{real}}^2 P_t(1 - P_t)\,\mathrm{d}t \cdot \left(1 - \frac{\sigma_{\mathrm{impl}}^2}{\sigma_{\mathrm{real}}^2}\right), \tag{70}$$

where Q is lot volume, σ_{impl} is the implied vol at which the market maker traded, and σ_{real} is realized vol. If the market maker prices at fair vol, the second term vanishes and P&L equals the spread capture.

13.2 Illustrative Economics

Assume the following scenario for a moderately active OIEC market:

- Underlying event future: 50,000 lots/day, 2-cent bid-ask spread.
- OIEC volume: 5,000 lots/day (10% of underlying), which is conservative for a newly listed options market.
- Average OIEC spread: \$0.005 per contract (\$5.00 per lot).
- Contract multiplier: \$1,000.

Table 4: Illustrative market-making economics for a single event future with listed OIECs.

Item	**Value**	**Notes**
Daily OIEC volume	5,000 lots	10% of underlying
Spread capture per lot	\$5.00	\$0.005 × \$1,000
Gross daily spread revenue	\$25,000	
Gamma hedging cost (realized ≈ implied)	−\$5,000	Discrete rebalancing slippage
Exchange and clearing fees	−\$2,500	\$0.50/lot assumed
Net daily P&L	**\$17,500**	
Net annualized (250 trading days)	\$4.375M	Per event future
Capital deployed (margin + inventory)	∼\$500K	
Return on capital	**∼875%**	Pre-tax, pre-overhead

These are aggressive but not unreasonable numbers for a new options market with limited competition. As more market makers enter, spreads compress and returns normalize. The key point is that the economics are strongly positive at launch.

13.3 Delta Hedging Feasibility

Market makers hedge in the underlying event future. Key considerations:

- The underlying must be liquid enough to absorb delta-hedging flow. If OIEC open interest reaches 100,000 lots with average delta 0.50, the outstanding hedge demand is 50,000 lots of the event future, comparable to one day's volume. This is manageable.
- Near boundaries ($P_t < 0.05$ or $P_t > 0.95$), gamma increases relative to delta, making hedging more expensive. Position limits on market makers near boundaries may be appropriate.
- Gamma bleeding from discrete rebalancing is approximately $\frac{1}{2}\Gamma\sigma^2 P(1-P)\Delta t$ per rebalance interval. At ATM with $\Gamma = 0.9$, $\sigma = 1.5$, $P = 0.5$, hourly rebalancing ($\Delta t = 1/6720$), the cost is roughly 0.015 cents per contract per hour.

13.4 Liquidity Bootstrapping

The typical chicken-and-egg problem (no liquidity without market makers, no market makers without liquidity) can be addressed through:

1. **Designated market maker (DMM) obligations:** Require DMMs to continuously quote a minimum number of strikes with maximum spread, in exchange for fee rebates or exclusive access to certain order types.
2. **Fee holidays:** Zero exchange fees for the first 6–12 months to reduce the break-even volume threshold.
3. **Seeded liquidity:** The exchange or affiliated entities provide initial quotes, absorbing early losses as a market development cost.
4. **Cross-margining with the underlying:** Allowing portfolio margin between event futures and OIECs reduces capital requirements for hedged positions, making market-making more capital-efficient from day one.

Part II
Advanced Extensions and Novel Frameworks

The preceding sections established the core OIEC framework: pricing, hedging, completeness, and exchange implementation. In this part, we develop novel extensions that push prediction market derivatives into genuinely new territory. Each section introduces mathematical structures, contract designs, or trading paradigms that have no direct precedent in existing financial markets.

14 The Belief Volatility Index (BVIX)

We construct a VIX-like index for prediction markets: a model-free, real-time measure of the market's expectation of future belief volatility, computed from the OIEC option strip.

14.1 Definition and Construction

Definition 14.1 (Belief Volatility Index)**.** For an event future with price P_t, the *Belief Volatility Index* (BVIX) at time t with horizon τ is

$$\mathrm{BVIX}_t^2(\tau) := \frac{1}{\tau - t}\mathbb{E}^{\mathbb{Q}}\left[\int_t^\tau \sigma_s^2 P_s(1-P_s)\,\mathrm{d}s \,\middle|\, \mathcal{F}_t\right], \tag{71}$$

the annualized expected realized belief variance over $[t, \tau]$.

Theorem 14.2 (Model-Free BVIX Formula)**.** *The BVIX can be computed model-free from observed OIEC option prices:*

$$\mathrm{BVIX}_t^2(\tau) = \frac{2e^{r(\tau-t)}}{\tau - t}\left[\int_0^{P^*} \frac{\mathfrak{p}(t;\tau,K)}{K^2(1-K)^2}\,\mathrm{d}K + \int_{P^*}^1 \frac{c(t;\tau,K)}{K^2(1-K)^2}\,\mathrm{d}K\right], \tag{72}$$

where $P^* = P_t$ is the current event future price, $\mathfrak{p}$ and c are observed put and call prices, and the weighting $K^{-2}(1-K)^{-2} = L''(K)$ comes from the second derivative of the bounded log payoff.

Proof. From the variance swap replication (Theorem 4.12), the expected quadratic variation equals the risk-neutral expectation of the bounded log payoff $L(P_\tau) - L(P_t)$, adjusted for the dynamic hedge. Taking the $\mathbb{Q}$-expectation:

$$\mathbb{E}^{\mathbb{Q}}\left[\int_t^\tau \sigma_s^2 P_s(1-P_s)\,\mathrm{d}s \,\middle|\, \mathcal{F}_t\right] = e^{r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[L(P_\tau) \mid \mathcal{F}_t] - L(P_t) + (\text{hedge PV}).$$

Apply the Carr–Madan static replication (Remark 4.13) to $\mathbb{E}^{\mathbb{Q}}[L(P_\tau) \mid \mathcal{F}_t]$:

$$\begin{aligned} e^{r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[L(P_\tau) \mid \mathcal{F}_t] = e^{r(\tau-t)}L(P^*) &+ e^{r(\tau-t)}L'(P^*)\underbrace{\mathbb{E}^{\mathbb{Q}}[P_\tau - P^* \mid \mathcal{F}_t]}_{=0} \\ &+ \int_0^{P^*} L''(K)\cdot e^{r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[(K-P_\tau)^+]\,\mathrm{d}K \\ &+ \int_{P^*}^1 L''(K)\cdot e^{r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[(P_\tau-K)^+]\,\mathrm{d}K. \end{aligned}$$

The linear term vanishes by the martingale property. The remaining integrals involve $e^{r(\tau-t)}$ times discounted put and call prices, which equal the undiscounted market prices. Substituting $L''(K) = K^{-2} + (1-K)^{-2}$ and dividing by $(\tau - t)$ gives (72). □

Remark 14.3 (Interpretation and Practical Use)**.** The BVIX is the prediction market analog of the CBOE VIX. A BVIX of 1.50 on an election contract means the market expects annualized probability volatility of 1.50 in the Jacobi sense, implying a daily standard deviation of $\sigma\sqrt{P(1-P)/252}$. At $P = 0.50$, this gives a daily move of ≈ 4.7 percentage points. The BVIX rises before debates, FOMC meetings, and election days, and collapses after event resolution. It can be computed in real time from the OIEC option strip without any model assumptions.

14.2 BVIX Futures and Options

Definition 14.4 (BVIX Futures)**.** A *BVIX future* with maturity τ_1 and measurement horizon $[\tau_1, \tau_2]$ settles at $\mathrm{BVIX}_{\tau_1}(\tau_2)$. The time-$t$ price is

$$F_t^{\mathrm{BVIX}} = \mathbb{E}^{\mathbb{Q}}[\mathrm{BVIX}_{\tau_1}(\tau_2) \mid \mathcal{F}_t]. \tag{73}$$

Definition 14.5 (BVIX Options)**.** A call option on the BVIX with strike K_v and maturity τ_1 pays $(\mathrm{BVIX}_{\tau_1} - K_v)^+$. This is a second-order derivative: an option on the volatility of a probability.

Proposition 14.6 (BVIX Contango)**.** *The BVIX future price exceeds the current BVIX spot level when belief volatility is itself volatile. Formally, if* BVIX_{τ_1} *is a non-degenerate random variable, then by Jensen's inequality and the convexity of the square root:*

$$F_t^{\mathrm{BVIX}} = \mathbb{E}^{\mathbb{Q}}\left[\sqrt{\mathrm{BVIX}_{\tau_1}^2} \,\middle|\, \mathcal{F}_t\right] \geq \sqrt{\mathbb{E}^{\mathbb{Q}}[\mathrm{BVIX}_{\tau_1}^2 \mid \mathcal{F}_t]} \geq \mathrm{BVIX}_t, \tag{74}$$

where the second inequality uses Jensen on the concave square root applied to $\mathbb{E}[\mathrm{BVIX}^2]$*, combined with the fact that vol-of-vol creates a convexity premium. This creates a natural term structure of BVIX futures that is typically in contango, analogous to the VIX futures term structure.*

Remark 14.7 (BVIX as a Product)**.** A listed BVIX future would be the first tradeable "fear gauge" for event uncertainty. Portfolio managers could buy BVIX calls as tail hedges against regime shifts in prediction markets. Event-driven hedge funds could go short BVIX after elections resolve (vol crush). The BVIX term structure would reveal the market's expectation of when uncertainty will peak (e.g., around debate dates vs. election day).

15 Path-Dependent Regime Contracts

We introduce contracts whose payoffs depend on the *trajectory* of the probability process, not merely its terminal value. These have no natural analog in equity markets because equities lack bounded state spaces with binary terminal conditions.

15.1 Probability Corridor Contracts

Definition 15.1 (Corridor Bond)**.** A *corridor bond* with boundaries $[a, b] \subset (0, 1)$ and maturity τ pays

$$\mathrm{CB}_{a,b,\tau} := \frac{1}{\tau}\int_0^\tau \mathbf{1}_{\{P_s \in [a,b]\}}\,\mathrm{d}s, \tag{75}$$

the fraction of time the probability process spends inside the corridor.

Theorem 15.2 (Corridor Bond Pricing)**.** *Under the martingale Jacobi model, the time-*0 *price of a corridor bond is*

$$\mathrm{CB}_0 = \frac{e^{-r\tau}}{\tau}\int_0^\tau \int_a^b p(0, P_0; s, y)\,\mathrm{d}y\,\mathrm{d}s, \tag{76}$$

where p *is the Jacobi transition density from Theorem 2.11.*

Proof. By Fubini's theorem and risk-neutral pricing:

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{1}{\tau}\int_0^\tau \mathbf{1}_{\{P_s\in[a,b]\}}\,\mathrm{d}s\right] = \frac{1}{\tau}\int_0^\tau \mathbb{Q}(P_s\in[a,b]\mid P_0)\,\mathrm{d}s = \frac{1}{\tau}\int_0^\tau\int_a^b p(0,P_0;s,y)\,\mathrm{d}y\,\mathrm{d}s.$$

□

Example 15.3 (Stability Contract)**.** A trader who believes the election probability will remain between 0.45 and 0.55 (a "coin-flip stays a coin-flip" view) can buy a corridor bond with $[a,b] = [0.45, 0.55]$. The contract pays close to 1 if the market stays undecided and close to 0 if it swings decisively.

15.2 Maximum Drawdown Contracts

Definition 15.4 (Belief Drawdown)**.** Define the running maximum $\bar{P}_t := \sup_{0\le s\le t} P_s$, the drawdown process $D_t := \bar{P}_t - P_t$, and the maximum drawdown $D^*_\tau := \sup_{0\le t\le\tau} D_t$.

Definition 15.5 (Drawdown Swap)**.** A *drawdown swap* pays $D^*_\tau - K_{\mathrm{DD}}$ at maturity, where K_{DD} is the strike.

Proposition 15.6 (Drawdown Bounds on the Unit Interval)**.** *For the Jacobi process on* $[0,1]$*:*

(i) $D^*_\tau \in [0,1]$*, with* $D^*_\tau = 1$ *only if the process reaches both 1 and 0.*

(ii) For short horizons, the fair strike satisfies

$$K^*_{\mathrm{DD}} = \mathbb{E}^{\mathbb{Q}}[D^*_\tau] \approx c\cdot\sigma\sqrt{P_0(1-P_0)\tau}, \quad c\approx\sqrt{\pi/2}\approx 1.25, \tag{77}$$

using the expected maximum of a locally Brownian process.

(iii) For long horizons, $K^*_{\mathrm{DD}} \to \min(P_0, 1-P_0)$ *as* $\tau\to\infty$ *under the martingale measure, since the process eventually reaches both boundaries.*

Proof. (i) is immediate from $P_t \in [0,1]$. For (ii), over short horizons the Jacobi process is approximately $P_t \approx P_0 + \sigma\sqrt{P_0(1-P_0)}W_t$, and the expected maximum drawdown of Brownian motion with volatility v over time τ is $v\sqrt{\pi\tau/2}$. Part (iii) follows because the martingale Jacobi process with $\alpha=\beta=1$ reaches both boundaries in finite time a.s. □

15.3 First-Passage Contracts

Definition 15.7 (First-Passage Digital)**.** For barrier $H\in(0,1)$, define the first-passage time $\tau_H := \inf\{t\ge 0 : P_t = H\}$. A *first-passage digital* pays $e^{-r\tau_H}\mathbf{1}_{\{\tau_H\le\tau\}}$.

Theorem 15.8 (First-Passage Probability via Spectral Expansion)**.** *Under the martingale Jacobi model, the probability that* P_t *reaches* $H > P_0$ *before time* τ *is*

$$\mathbb{Q}(\tau_H\le\tau\mid P_0) = 1 - \sum_{n=0}^{\infty} a_n(P_0,H)\,e^{-\mu_n\tau}, \tag{78}$$

where μ_n *and* a_n *are the eigenvalues and coefficients of the Jacobi generator* killed *at* H*, i.e., restricted to* $[0,H)$ *with absorbing boundary at* H*. The killed eigenvalues satisfy* $\mu_n > \lambda_n$ *(faster decay than the unkilled process), reflecting the loss of probability mass at the absorbing barrier.*

Proof. Let $u(t,x) := \mathbb{Q}(\tau_H > t \mid P_0 = x)$ be the survival probability. It satisfies

$$\frac{\partial u}{\partial t} = \frac{\sigma^2}{2}x(1-x)\frac{\partial^2 u}{\partial x^2}, \quad u(0,x)=1 \text{ for } x\in[0,H), \quad u(t,H)=0.$$

Expanding in the eigenfunctions of $\mathcal{L}$ on $[0,H)$ with Dirichlet condition at H gives (78). □

Remark 15.9 (Trading Reachability)**.** First-passage contracts let traders bet on "will the probability ever reach 0.80?" This is fundamentally different from a European call (which requires $P_\tau \geq 0.80$ *at expiry*) and from a one-touch barrier. It captures the market's view on whether enough information will arrive to push the probability to a given level at *any* point, regardless of subsequent reversals.

16 Probability-Weighted Portfolio Theory

We develop the analog of Markowitz mean-variance optimization when the "assets" are event futures on $[0, 1]$, deriving a state-dependent efficient frontier and new separation results.

16.1 The Event Portfolio Problem

Definition 16.1 (Event Portfolio)**.** An *event portfolio* consists of positions $w = (w_1, \ldots, w_n) \in \mathbb{R}^n$ in n event futures with prices $P_t = (P_t^{(1)}, \ldots, P_t^{(n)}) \in [0, 1]^n$. The portfolio return over $[t, t + \Delta t]$ is

$$R_t = \sum_{i=1}^{n} w_i \Delta P_t^{(i)}. \tag{79}$$

Theorem 16.2 (State-Dependent Covariance and Efficient Frontier)**.** *Under the multivariate Jacobi model (Definition 7.5 generalized to n assets), the instantaneous covariance matrix is*

$$\Sigma_{ij}(P_t) = \sigma_i \sigma_j \rho_{ij} \sqrt{P_t^{(i)}(1 - P_t^{(i)})} \sqrt{P_t^{(j)}(1 - P_t^{(j)})}, \tag{80}$$

which is state-dependent*: it changes with the probability levels. The mean-variance efficient frontier at time t solves*

$$\min_{w \in \mathbb{R}^n} \; w^\top \Sigma(P_t)\, w \quad \textit{subject to} \quad w^\top \mu = \bar{\mu}, \quad w^\top \mathbf{1} = W, \tag{81}$$

where $\mu = (\mu_1, \ldots, \mu_n)$ is the vector of expected returns under $\mathbb{P}$ and W is total capital.

Proof. The covariance follows from the bivariate dynamics:

$$\mathrm{Cov}(\Delta P^{(i)}, \Delta P^{(j)}) = \sigma_i \sigma_j \rho_{ij} \sqrt{P^{(i)}(1 - P^{(i)})} \sqrt{P^{(j)}(1 - P^{(j)})} \Delta t$$

The optimization is standard quadratic programming with linear constraints. □

Proposition 16.3 (Vanishing Diversification at Certainty)**.** *As any event future $P_t^{(i)} \to 0$ or $P_t^{(i)} \to 1$:*

(i) $\Sigma_{ij}(P_t) \to 0$ for all j: the event's contribution to portfolio risk vanishes.

(ii) The optimal weight w_i^* degenerates to pure carry (holding for the expected settlement value) with zero risk contribution.

(iii) Diversification benefit from event i is maximized when $P_t^{(i)} = 1/2$.

Proof. (i) follows from $\sqrt{P^{(i)}(1 - P^{(i)})} \to 0$ as $P^{(i)} \to 0$ or 1. For (ii), when $\Sigma_{ij} = 0$, the risk constraint is not binding and the optimal allocation treats event i as a riskless asset with deterministic payoff. Part (iii) follows from $\max_{p \in [0,1]} p(1 - p) = 1/4$ at $p = 1/2$. □

Corollary 16.4 (Maximum Uncertainty Portfolio)**.** *The portfolio with maximum variance ("maximum uncertainty" or "maximum diversification potential") is concentrated in events near $P^{(i)} = 1/2$. The tangency portfolio of prediction markets overweights uncertain events relative to near-certain ones, all else being equal. This is the opposite of a cap-weighted equity portfolio, where the largest (most certain) positions dominate.*

Proposition 16.5 (Dynamic Rebalancing Rule)**.** *As event futures move toward resolution ($P^{(i)} \to 0$ or 1), the covariance matrix $\Sigma(P_t)$ changes, requiring portfolio rebalancing. The optimal rebalancing frequency is state-dependent: rebalance more frequently when multiple events are near $P = 1/2$ (maximum volatility) and less frequently when events are near resolution.*

17 Continuous-Time Prediction Markets via Jacobi Bonding Curves

Current prediction markets operate through discrete limit order books. We propose a fundamentally different architecture: a *bonding curve* that mechanically maps aggregate demand to the probability price, where the curve itself is designed so that the resulting price process satisfies the Jacobi dynamics by construction. This merges automated market maker (AMM) design with the stochastic calculus framework of Part I.

17.1 The Jacobi Bonding Curve

Definition 17.1 (Jacobi Bonding Curve AMM)**.** A *Jacobi bonding curve* is defined by a price-impact function $g : (0, 1) \to \mathbb{R}_+$ and a liquidity depth parameter $\ell > 0$ such that:

(i) The event future price is $P_t \in (0, 1)$, updated by incoming trades.

(ii) The instantaneous price impact of a trade of size $\mathrm{d}q$ (signed lot volume) is

$$\mathrm{d}P = g(P)\,\mathrm{d}q, \quad g(P) := \frac{\sigma}{\sigma_q}\sqrt{P(1-P)}, \tag{82}$$

where $\sigma_q > 0$ is the volatility of the order flow process.

(iii) The cost to a trader moving the price from P_a to P_b is

$$C(P_a, P_b) = \int_{P_a}^{P_b} \frac{u}{g(u)}\,\mathrm{d}u = \frac{\sigma_q}{\sigma}\int_{P_a}^{P_b} \frac{u}{\sqrt{u(1-u)}}\,\mathrm{d}u = \frac{\sigma_q}{\sigma}\left[-\sqrt{1-P}\right]_{P_a}^{P_b} + (\text{forward price adj.}). \tag{83}$$

The key design choice in (82) is that $g(P) \propto \sqrt{P(1-P)}$: price impact vanishes at the boundaries (it takes overwhelming evidence to move a near-certain probability) and is maximized at $P = 1/2$ (maximum uncertainty means maximum sensitivity to new information).

Theorem 17.2 (Jacobi Dynamics from Order Flow)**.** *Suppose net demand q_t follows an arithmetic Brownian motion $\mathrm{d}q_t = \mu_q\,\mathrm{d}t + \sigma_q\,\mathrm{d}W_t$ under the physical measure. Then the price process P_t induced by the bonding curve* (82) *satisfies*

$$\mathrm{d}P_t = g(P_t)\mu_q\,\mathrm{d}t + \frac{\sigma^2(1-2P_t)}{4}\,\mathrm{d}t + \sigma\sqrt{P_t(1-P_t)}\,\mathrm{d}W_t. \tag{84}$$

Under the risk-neutral measure $\mathbb{Q}$ (absorbing both drift terms into the market price of risk), this becomes

$$\mathrm{d}P_t = \sigma\sqrt{P_t(1-P_t)}\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{85}$$

recovering exactly the martingale Jacobi diffusion (6)*.*

Proof. Apply Itô's lemma to $P_t = P(q_t)$, where $P(\cdot)$ is the inverse of the cumulative impact function $Q(P) = \int^P g(u)^{-1}\,\mathrm{d}u$:

$$\mathrm{d}P_t = P'(q_t)\,\mathrm{d}q_t + \tfrac{1}{2}P''(q_t)\sigma_q^2\,\mathrm{d}t.$$

Since $P'(q) = g(P(q))$, the diffusion term is $g(P_t)\sigma_q = \sigma\sqrt{P_t(1-P_t)}$. For the second-order correction, $P''(q) = g'(P)g(P)$, yielding

$$\tfrac{1}{2}P''(q_t)\sigma_q^2 = \tfrac{1}{2}g'(P_t)g(P_t)\sigma_q^2 = \frac{\sigma^2(1-2P_t)}{4}.$$

This drift is deterministic given P_t and can be absorbed into the Girsanov change of measure (Remark 2.9). □

Definition 17.3 (Reserve Function)**.** The bonding curve AMM maintains reserves $R(P)$ satisfying

$$R(P) = R_0 + \int_{P_0}^{P} \frac{u}{g(u)}\,\mathrm{d}u, \tag{86}$$

which is the cumulative cost of moving the price from P_0 to P. For the Jacobi bonding curve, this integral has a closed form involving $\arcsin(\sqrt{P})$.

Proposition 17.4 (Liquidity Provider Impermanent Loss)**.** *A liquidity provider who seeds the bonding curve at price P_0 faces impermanent loss (IL) when the price moves to P_T:*

$$\mathrm{IL}(P_0, P_T) = R(P_T) - R(P_0) - P_T(Q(P_T) - Q(P_0)), \tag{87}$$

where $Q(P) = \int^P g(u)^{-1}\,\mathrm{d}u$ is the cumulative demand. This IL is bounded:

$$|\mathrm{IL}| \le \ell \cdot |L(P_T) - L(P_0)|, \tag{88}$$

where $L(P) = -\log P - \log(1-P)$ is the bounded log payoff from (37)*. The impermanent loss of the AMM liquidity provider is directly linked to the belief variance swap payoff.*

Remark 17.5 (Advantages Over Limit Order Books)**.** The bonding curve approach offers structural advantages for prediction markets: (i) continuous liquidity at all price levels without requiring market makers, (ii) the price process has known analytical properties by construction, so OIECs can be priced using the exact Jacobi transition density, (iii) the curve parameters (σ, ℓ) are directly observable, making calibration trivial, and (iv) impermanent loss, the main cost of liquidity provision, is analytically tractable and hedgeable via belief variance swaps.

18 Automated Market Making with Provable Greeks Bounds

We design a market-making algorithm that uses the OIEC pricing PDE to guarantee bounded risk exposure at all times.

18.1 The Greeks-Bounded Market Maker

Definition 18.1 (GBMM Protocol)**.** A *Greeks-Bounded Market Maker* (GBMM) operates with risk limits $(\bar{\Delta}, \bar{\Gamma}, \bar{\mathcal{V}}) \in \mathbb{R}_+^3$. At each time t, the GBMM:

(i) Computes portfolio Greeks $(\Delta_t^{\mathrm{port}}, \Gamma_t^{\mathrm{port}}, \mathcal{V}_t^{\mathrm{port}})$ from all outstanding positions using the pricing PDE (52).

(ii) For each incoming order with Greeks $(\delta\Delta, \delta\Gamma, \delta\mathcal{V})$, checks feasibility:

$$|\Delta_t^{\mathrm{port}} + \delta\Delta| \le \bar{\Delta}, \quad |\Gamma_t^{\mathrm{port}} + \delta\Gamma| \le \bar{\Gamma}, \quad |\mathcal{V}_t^{\mathrm{port}} + \delta\mathcal{V}| \le \bar{\mathcal{V}}. \tag{89}$$

(iii) Quotes spreads that widen as limits are approached:

$$s_t = s_0 \cdot \left(1 + \alpha \max\left(\frac{|\Delta_t^{\mathrm{port}}|}{\bar{\Delta}}, \frac{|\Gamma_t^{\mathrm{port}}|}{\bar{\Gamma}}, \frac{|\mathcal{V}_t^{\mathrm{port}}|}{\bar{\mathcal{V}}}\right)^2\right), \tag{90}$$

where s_0 is the base spread and $\alpha > 0$ controls the widening rate.

(iv) Rejects orders that would violate (89).

Theorem 18.2 (Deterministic Loss Bound)**.** *Under the GBMM protocol with limits* $(\bar{\Delta}, \bar{\Gamma}, \bar{\mathcal{V}})$*, the maximum portfolio loss over any single time step* Δt *is bounded by*

$$|\Delta V^{\text{port}}| \leq \bar{\Delta} \cdot \Delta P^{\max} + \tfrac{1}{2}\bar{\Gamma} \cdot (\Delta P^{\max})^2 + \bar{\mathcal{V}} \cdot \Delta\sigma^{\max} + (r \cdot V^{\max} + \tfrac{1}{4}\bar{\Gamma}\sigma^2) \cdot \Delta t, \tag{91}$$

where $\Delta P^{\max}$ *and* $\Delta\sigma^{\max}$ *are the maximum price and vol moves (from margin scan ranges), and* $V^{\max}$ *is the maximum portfolio value. This bound is* deterministic *and holds path-by-path.*

Proof. By Taylor expansion of the portfolio value $V^{\text{port}}(t + \Delta t, P_{t+\Delta t}, \sigma_{t+\Delta t})$:

$$\Delta V^{\text{port}} \approx \Delta^{\text{port}} \Delta P + \tfrac{1}{2}\Gamma^{\text{port}}(\Delta P)^2 + \mathcal{V}^{\text{port}} \Delta\sigma + \Theta^{\text{port}} \Delta t.$$

By the theta–gamma identity (46), $|\Theta^{\text{port}}| \leq |r|V^{\max} + \frac{1}{2}|\Gamma^{\text{port}}|\sigma^2 \cdot \max_{P\in[0,1]} P(1-P)$. Since $P(1-P) \leq 1/4$ and the constraints (89) bound all Greeks, taking absolute values gives (91). □

Proposition 18.3 (Spread Revenue Lower Bound)**.** *If the GBMM fills volume* Q *per period at average spread* $\bar{s} \geq s_0$*, the net P&L satisfies*

$$P\&L \geq Q \cdot s_0 - \left(\bar{\Delta} \cdot \Delta P^{\max} + \tfrac{1}{2}\bar{\Gamma}(\Delta P^{\max})^2 + \bar{\mathcal{V}} \cdot \Delta\sigma^{\max}\right) \cdot N_{\text{rebal}}, \tag{92}$$

where N_{rebal} *is the number of rebalancing periods. The GBMM is profitable when volume exceeds a deterministic break-even threshold.*

Remark 18.4 (Regulatory and DeFi Applications)**.** The GBMM provides *provable* risk bounds—at no point can the algorithm accumulate Greeks exposure beyond the specified limits. This makes it suitable for (i) regulated exchanges where risk limits must be demonstrably enforced, (ii) decentralized finance (DeFi) protocols where trust is replaced by mathematical guarantees, and (iii) internal risk management at market-making firms as a hard-coded backstop.

19 Self-Referential Prediction Derivatives

We introduce contracts whose payoffs depend on the *derivative market's own pricing*, creating recursive structures unique to prediction markets.

19.1 Implied Volatility Options

Definition 19.1 (Implied Vol Call)**.** An *implied volatility call* with strike K_σ and maturity τ_1 pays

$$(\sigma^{\text{imp}}_{\tau_1}(K_0, \tau_2) - K_\sigma)^+, \tag{93}$$

where $\sigma^{\text{imp}}_{\tau_1}(K_0, \tau_2)$ is the at-the-money implied belief volatility at time τ_1 for OIECs expiring at $\tau_2 > \tau_1$.

Theorem 19.2 (Existence of Equilibrium Price via Fixed-Point)**.** *The price of an implied volatility call requires solving a fixed-point problem. Let* V *denote the IV call price. The implied volatility* $\sigma^{\text{imp}}_{\tau_1}$ *depends on OIEC prices at* τ_1*, which depend on the volatility process, which may depend on* V *through feedback effects. Define the pricing map*

$$\Phi(V) := e^{-r(\tau_1 - t)}\mathbb{E}^{\mathbb{Q}}\left[(\sigma^{\text{imp}}_{\tau_1}(\omega, V) - K_\sigma)^+ \mid \mathcal{F}_t\right]. \tag{94}$$

If $\Phi : [0, \bar{V}] \to [0, \bar{V}]$ *is continuous, then by the Brouwer fixed-point theorem, there exists* $V^* = \Phi(V^*)$.

Proof. The domain $[0, \bar{V}]$ is compact and convex (a closed interval in $\mathbb{R}$). The map Φ is continuous because: (a) σ^{imp} is continuous in option prices by the implicit function theorem (vega is strictly positive), (b) option prices are continuous in the volatility process, and (c) the expectation operator preserves continuity under dominated convergence. By Brouwer's theorem, Φ has a fixed point.

The upper bound $\bar{V} = e^{-r(\tau_1 - t)}(\bar{\sigma} - K_\sigma)^+$ exists because implied volatility is bounded above by $\bar{\sigma}$ (since the underlying is bounded in $[0, 1]$, the implied vol cannot exceed the level at which the call price equals 1). □

Remark 19.3 (Uniqueness and Multiplicity)**.** Uniqueness of the fixed point is not guaranteed. Multiple equilibria may exist if the feedback is strong: the IV call being expensive causes market makers to widen spreads, raising implied vol, which raises the IV call price. This creates the possibility of "self-fulfilling volatility expectations" that have no analog in standard equity derivatives. Investigating conditions for uniqueness (e.g., contractivity of Φ) is an open question.

19.2 Calibration Score Contracts

Definition 19.4 (Calibration Score)**.** A *calibration score contract* with event maturity T pays

$$\mathrm{CS}_T := 1 - (X_T - P_{T^-})^2, \tag{95}$$

where $X_T \in \{0, 1\}$ is the event outcome and P_{T^-} is the pre-terminal market probability. This equals 1 when the market was perfectly calibrated ($P_{T^-} = X_T$) and is minimized when the market was maximally wrong.

Proposition 19.5 (Calibration Score Fair Value)**.** *Under the model $\mathbb{Q}(E \mid P_{T^-}) = P_{T^-}$ (the pre-terminal probability is well-calibrated):*

$$\mathbb{E}^{\mathbb{Q}}[\mathrm{CS}_T \mid \mathcal{F}_t] = 1 - \mathbb{E}^{\mathbb{Q}}[P_{T^-}(1 - P_{T^-}) \mid \mathcal{F}_t]. \tag{96}$$

Proof. Compute $\mathbb{E}[(X_T - P_{T^-})^2 \mid P_{T^-}]$:

$$\mathbb{E}[X_T^2 \mid P_{T^-}] - 2P_{T^-}\mathbb{E}[X_T \mid P_{T^-}] + P_{T^-}^2 = P_{T^-} - 2P_{T^-}^2 + P_{T^-}^2 = P_{T^-}(1 - P_{T^-}).$$

Taking the outer expectation and using iterated expectations gives (96). □

Corollary 19.6 (Calibration–Variance Duality)**.** *The calibration score is inversely related to the belief variance swap:*

$$\mathbb{E}[\mathrm{CS}_T] = 1 - \frac{K^*_{\mathrm{var},T}}{\sigma^2(T - t)}, \tag{97}$$

where $K^*_{\mathrm{var},T}$ is the variance swap strike with maturity T. Markets with high expected belief variance (high BVIX) have lower expected calibration scores: more uncertain markets are less well-calibrated.

20 Maximum Entropy Pricing on $[0, 1]$

We derive a pricing framework that replaces the Jacobi diffusion assumption with a maximum-entropy principle, yielding a model-free, nonparametric approach that nests the Jacobi model as a special case.

20.1 The Maximum Entropy Principle on $[0, 1]$

Definition 20.1 (Maximum Entropy Pricing Measure)**.** Given observed OIEC prices $\{c_k^{\text{mkt}}\}_{k=1}^m$ for calls at strikes $\{K_k\}$ and maturities $\{\tau_k\}$, the *maximum entropy pricing measure* is

$$\mathbb{Q}^* = \arg\max_{\mathbb{Q}\in\mathcal{M}} H(\mathbb{Q}), \quad H(\mathbb{Q}) := -\int_0^1 f(y)\log f(y)\,\mathrm{d}y, \tag{98}$$

where $\mathcal{M}$ is the set of probability densities f on $[0, 1]$ satisfying:

(i) Martingale constraint: $\int_0^1 y\, f(y)\,\mathrm{d}y = P_t$.

(ii) Pricing constraints: $\int_0^1 (y - K_k)^+ f(y)\,\mathrm{d}y = e^{r(\tau_k - t)} c_k^{\text{mkt}}$ for all k.

(iii) Support: $f(y) > 0$ for $y \in (0, 1)$, $f(y) = 0$ outside.

Theorem 20.2 (Maximum Entropy Solution: Generalized Beta)**.** *The maximum entropy solution on* $[0, 1]$ *subject to a mean constraint* $\mathbb{E}[Y] = \mu$ *is the Beta distribution:*

$$f^*(y) = \frac{y^{\alpha^*-1}(1-y)^{\beta^*-1}}{B(\alpha^*, \beta^*)}, \quad \alpha^* = \mu\phi, \quad \beta^* = (1-\mu)\phi, \tag{99}$$

where $\phi > 0$ *is a concentration parameter. With additional constraints (pricing constraints from observed options), the solution is a member of the exponential family:*

$$f^*(y) = \frac{1}{Z}\exp\left(\sum_{k=1}^m \lambda_k (y - K_k)^+\right) \cdot y^{\alpha^*-1}(1-y)^{\beta^*-1}, \tag{100}$$

where $\lambda_1, \ldots, \lambda_m$ *are Lagrange multipliers determined by the pricing constraints and* Z *is a normalizing constant.*

Proof. The maximum entropy problem is a convex optimization over probability densities on $[0, 1]$. By the method of Lagrange multipliers, the optimality condition is

$$-\log f^*(y) - 1 + \lambda_0 + \lambda_\mu y + \sum_{k=1}^m \lambda_k (y - K_k)^+ = 0,$$

where λ_0 enforces normalization, λ_μ enforces the mean constraint, and λ_k enforce the pricing constraints. Solving for f^*:

$$f^*(y) = \exp\left(\lambda_0 - 1 + \lambda_\mu y + \sum_k \lambda_k (y - K_k)^+\right).$$

With no pricing constraints ($m = 0$), this reduces to $f^*(y) = C \cdot e^{\lambda_\mu y}$ on $[0, 1]$, which after reparameterization and normalization is the Beta distribution with mean μ. The concentration ϕ is determined by the mean constraint alone when no additional constraints are present; with pricing constraints, the Lagrange multipliers λ_k shape the distribution to match observed prices. □

Corollary 20.3 (Nesting the Jacobi Model)**.** *The Jacobi transition density at time* τ *is approximately* $\text{Beta}(\alpha(\tau), \beta(\tau))$ *for certain parameter values. The maximum entropy framework nests the Jacobi model: if observed option prices are consistent with Jacobi dynamics, the entropy-maximizing measure recovers the Jacobi transition density. If they are not (e.g., due to jumps, stochastic vol, or microstructure effects), the entropy measure provides the* least biased *alternative consistent with the data.*

20.2 Pricing Exotic Claims

Proposition 20.4 (Entropy-Based Exotic Pricing)**.** *Given the maximum entropy density f^*, any claim $H(P_\tau)$ is priced as*

$$V_t^{\mathrm{ME}} = e^{-r(\tau-t)} \int_0^1 H(y)\, f^*(y)\, \mathrm{d}y. \tag{101}$$

For the unconstrained Beta case, many exotic claims have closed forms. For example:

(i) Corridor bond: $V = e^{-r\tau}[I_b(\alpha^*, \beta^*) - I_a(\alpha^*, \beta^*)]$, where I_x is the regularized incomplete beta function.

(ii) Digital call: $V = e^{-r\tau}[1 - I_K(\alpha^*, \beta^*)]$.

(iii) Variance swap strike: $V = \frac{\alpha^*\beta^*}{(\alpha^*+\beta^*)^2(\alpha^*+\beta^*+1)}$ (the variance of the Beta distribution).

Remark 20.5 (Model Risk Reduction)**.** The entropy approach eliminates the need to assume a specific diffusion model. If market prices of vanilla OIECs are inconsistent with the constant-σ Jacobi process, the entropy pricing measure automatically adapts. It is the "least presumptuous" measure consistent with observed data: it introduces no structure beyond what the data demand. For an exchange listing exotic OIECs (corridor bonds, drawdown swaps, first-passage contracts), entropy pricing provides a robust fallback when the Jacobi model is misspecified.

20.3 Connection to Relative Entropy and the Esscher Transform

Proposition 20.6 (Minimal Relative Entropy Pricing)**.** *The entropy pricing measure $\mathbb{Q}^*$ is also the solution to the dual problem*

$$\mathbb{Q}^* = \arg\min_{\mathbb{Q}\in\mathcal{M}} D_{\mathrm{KL}}(\mathbb{Q}\|\mathbb{Q}_0) = \arg\min_{\mathbb{Q}\in\mathcal{M}} \int_0^1 f(y) \log \frac{f(y)}{f_0(y)}\, \mathrm{d}y, \tag{102}$$

where $\mathbb{Q}_0$ is the uniform ("maximum ignorance") prior on $[0,1]$ and D_{KL} is the Kullback–Leibler divergence. This formulation connects to the Esscher transform in insurance mathematics and the minimal entropy martingale measure in incomplete market theory (Delbaen and Schachermayer, 1994).

21 Economic Applications

21.1 Hedging Political Risk

Consider a firm whose profits depend on a policy outcome E (e.g., tariff imposition).

Example 21.1 (Tariff Hedge)**.** A manufacturing firm faces costs increasing by \$10M if tariffs are imposed. To hedge:

1. Buy event futures on E with notional \$10M.
2. If E occurs ($X_T = 1$), the futures pay \$10M, offsetting the cost increase.
3. If E does not occur, the firm loses the premium but avoids the tariff cost.

The OIEC framework allows finer-grained hedging via puts and calls, enabling the firm to express views on probability ranges rather than binary outcomes.

21.2 Speculation on Probability Volatility

Traders may believe that market uncertainty about an event (not the event probability itself) is mispriced.

Example 21.2 (Volatility Trade)**.** If a trader believes election uncertainty is higher than implied by option prices:

1. Buy a probability straddle at $K = P_t$ (the current probability).
2. The straddle profits if P_τ moves away from K in either direction.
3. Alternatively, go long a belief variance swap.

21.3 Relative Value Across Events

With OIECs on multiple events, traders can express views on correlations.

Example 21.3 (Correlation Trade)**.** Two events E_1 (Fed hike) and E_2 (recession in 2 years) may be correlated. If a trader believes the correlation is higher than priced:

1. Buy a spread option paying $(P_\tau^{(2)} - P_\tau^{(1)})^+$.
2. If both events become more likely simultaneously (positive correlation), the spread tightens, and the option pays off.

21.4 Institutional Risk Management

Institutions with event-contingent exposures (banks with regulatory risk, energy firms with policy risk, asset managers with election sensitivity) can use OIECs to:

1. Express views on event probability ranges (not just binary outcomes).
2. Measure and report Greeks-based risk to internal risk committees.
3. Achieve targeted hedges using spreads and collars at known maximum cost.

22 Regulatory and Implementation Considerations

22.1 Exchange Infrastructure Requirements

For OIECs to function in practice, the underlying event futures must trade on regulated exchanges with:

- **Central clearing:** A central counterparty (CCP) to manage credit risk, with the margin framework from Section 12.
- **Margin requirements:** Daily variation margin to mark positions to market.
- **Position limits:** Caps on individual positions to prevent manipulation.
- **Objective settlement:** Clear, legally enforceable event definitions with settlement tied to official data releases (e.g., BLS inflation numbers, certified election results).

22.2 Regulatory Classification

In the United States, prediction markets and their derivatives face classification challenges:

- The CFTC has jurisdiction over futures and options on futures.
- Some prediction market contracts have been classified as gaming or gambling, restricting their availability.
- Recent rulings (e.g., CFTC approval of Kalshi political contracts) have expanded permissible event types.

OIECs, as derivatives on already-approved event futures, inherit the regulatory status of their underlyings.

22.3 Capital Treatment

For institutional users, OIECs must fit into existing capital frameworks:

- **Market risk:** OIECs contribute to Value-at-Risk (VaR) calculations via their delta, gamma, and vega exposures.
- **Counterparty credit risk:** Cleared contracts receive favorable capital treatment under Basel III. The bounded payoff structure ($\leq$ \$1,000 per lot) simplifies capital calculations relative to equity options with unbounded loss potential.
- **Operational risk:** Settlement based on external events introduces oracle risk.

22.4 Manipulation and Integrity

Prediction markets are vulnerable to manipulation if participants can influence the underlying event. For derivatives:

- OIECs amplify leverage, potentially increasing manipulation incentives.
- Mitigation includes position limits, surveillance, and diversification of settlement sources.
- Economic analysis suggests that deep, liquid markets are harder to manipulate profitably.

23 Conclusion

This paper has developed a comprehensive framework for Option-Implied Event Contracts, extending prediction market infrastructure into the domain of derivatives. Our main contributions are:

(1) **Mathematical foundation:** We established the Jacobi diffusion as a natural model for bounded probability dynamics and derived the pricing PDE, transition densities, and risk-neutral valuation formulas.

(2) **Contract definitions:** We introduced European and American probability options, confidence-weighted event digitals, and belief variance swaps, each serving distinct economic functions.

(3) **Pricing and hedging:** We derived closed-form pricing formulas, put-call parity relations, the theta–gamma identity, and a complete Greeks calculus enabling delta, gamma, and vega hedging.

(4) **Model-free variance replication:** We showed that belief variance swaps can be replicated by a static strip of probability options via the bounded log payoff, extending the Neuberger–Carr–Lee framework to $[0, 1]$.

(5) **Market completeness:** We proved that the OIEC market is complete, the risk-neutral measure is unique, and arbitrary bounded claims can be replicated.

(6) **Multi-event extensions:** We developed the bivariate Jacobi framework for correlated event futures, enabling pricing of spread and conditional options.

(7) **Numerical methods and calibration:** We presented trinomial trees, finite differences, and Monte Carlo algorithms, with a worked numerical example demonstrating calibration and pricing.

(8) **Exchange implementation:** We provided a prototype product specification (Section 11), margin methodology (Section 12), and market-making economics (Section 13) showing attractive returns at plausible volumes.

Beyond the core framework, we developed several advanced extensions: the Belief Volatility Index (BVIX), a model-free measure of event uncertainty computed from the option strip; path-dependent regime contracts including corridor bonds, drawdown swaps, and first-passage digitals; a probability-weighted portfolio theory with a state-dependent efficient frontier where diversification vanishes at certainty; Jacobi bonding curves that produce the Jacobi diffusion by construction from order flow; an automated market-making protocol (GBMM) with deterministic worst-case loss bounds; self-referential prediction derivatives whose pricing requires fixed-point existence proofs; and a maximum-entropy nonparametric pricing framework that nests the Jacobi model as a special case while providing a robust fallback when it is misspecified.

The bounded underlying is an advantage, not a complication: it caps losses, simplifies margin, and enables portfolio offsets not available in equity options. By providing hedging instruments, enabling institutional risk management, and creating liquidity linkages, OIECs have the potential to transform prediction markets from niche forecasting tools into integral components of global financial infrastructure.

Future research directions include:

- Empirical testing of the Jacobi model against historical prediction market data from platforms such as Kalshi and Polymarket.
- Development of richer multi-event correlation models, potentially building on the multivariate Jacobi framework of Gouriéroux and Jasiak (2006).
- Analysis of market microstructure under OIEC trading, including the feedback effects of delta hedging on the underlying event future.
- Extension to continuous-outcome events (e.g., GDP growth, inflation levels) rather than binary events.
- Detailed modeling of terminal information arrival for options expiring at or near event maturity, using Brownian bridge or hazard-rate approaches.
- Full treatment of the jump-diffusion Jacobi model (Remark 2.10), including the pricing of jump risk premia and the additional instruments required for completeness.
- Uniqueness conditions for self-referential fixed-point prices (Theorem 19.2) and analysis of equilibrium multiplicity.

A Proofs of Main Results

A.1 Proof of Theorem 2.11

The Jacobi diffusion is a classical example of a one-dimensional diffusion with polynomial eigenfunctions. The generator is

$$\mathcal{L}f(x) = \frac{\sigma^2}{2}x(1-x)f''(x).$$

The stationary measure (when it exists) is Beta(α, β) with density $w(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$. The eigenfunctions are Jacobi polynomials $P_n^{(\alpha-1,\beta-1)}(2x-1)$, orthogonal with respect to w. The eigenvalues are $\lambda_n = \frac{\sigma^2}{2}n(n+\alpha+\beta-1)$. The spectral expansion of the transition density follows from standard theory.

A.2 Derivation of the Greeks

Let $c(t, P) = e^{-r(\tau-t)}\mathbb{E}^{\mathbb{Q}}[(P_\tau - K)^+ \mid P_t = P]$. Differentiating under the integral:

$$\Delta_c = \frac{\partial c}{\partial P} = e^{-r(\tau-t)} \int_K^1 (y-K)\frac{\partial p(t, P; \tau, y)}{\partial P}\,\mathrm{d}y.$$

For the Jacobi process, the derivative of the transition density with respect to the initial condition can be computed from the eigenfunction expansion or by solving the adjoint PDE.

Gamma is the second derivative:

$$\Gamma_c = \frac{\partial^2 c}{\partial P^2} = e^{-r(\tau-t)} \int_K^1 (y-K)\frac{\partial^2 p}{\partial P^2}\,\mathrm{d}y.$$

Vega involves differentiating with respect to σ:

$$\mathcal{V}_c = \frac{\partial c}{\partial \sigma} = e^{-r(\tau-t)} \int_K^1 (y-K)\frac{\partial p}{\partial \sigma}\,\mathrm{d}y.$$

B Trinomial Tree Derivation

At node (n, k) with $P = k\Delta P$, the local variance of the Jacobi process over interval Δt is

$$\mathrm{Var}(P_{t+\Delta t} \mid P_t = P) \approx \sigma^2 P(1-P)\Delta t.$$

The trinomial tree has transitions to $P + \Delta P$ (up), P (middle), and $P - \Delta P$ (down) with probabilities (q_u, q_m, q_d). The moment-matching conditions are:

$$q_u + q_m + q_d = 1, \tag{103}$$

$$q_u\Delta P - q_d\Delta P = 0, \tag{104}$$

$$q_u(\Delta P)^2 + q_d(\Delta P)^2 = \sigma^2 P(1-P)\Delta t. \tag{105}$$

From (104): $q_u = q_d$. From (105): $2q_u(\Delta P)^2 = \sigma^2 P(1-P)\Delta t$, so

$$q_u = q_d = \frac{\sigma^2 P(1-P)\Delta t}{2(\Delta P)^2}.$$

From (103): $q_m = 1 - 2q_u$.

For $q_u, q_d, q_m \in (0, 1)$, we need $q_u < 1/2$, i.e., $\sigma^2 P(1-P)\Delta t < (\Delta P)^2$. Since $P(1-P) \leq 1/4$, this is satisfied for $\sigma^2\Delta t < 4(\Delta P)^2$.

At boundaries $k = 0$ or $k = M$, special treatment is needed: we use a binomial tree with $q_m = 1 - q$ and transition to the interior only.

C Finite Difference Implementation Details

Let $V_j^n = V(t_n, P_j)$ with $P_j = j\Delta P$ and $t_n = n\Delta t$. The Crank–Nicolson scheme for (52) is:

$$\frac{V_j^{n+1} - V_j^n}{\Delta t} = \frac{1}{2}(L^{n+1} + L^n)V_j,$$

where

$$LV_j = \frac{\sigma^2 P_j(1 - P_j)}{2(\Delta P)^2}(V_{j+1} - 2V_j + V_{j-1}) - rV_j.$$

Rearranging:

$$-a_j V_{j-1}^{n+1} + b_j V_j^{n+1} - c_j V_{j+1}^{n+1} = a_j V_{j-1}^n + d_j V_j^n + c_j V_{j+1}^n,$$

where

$$a_j = c_j = \frac{\sigma^2 P_j(1 - P_j)\Delta t}{4(\Delta P)^2}, \tag{106}$$

$$b_j = 1 + 2a_j + \frac{r\Delta t}{2}, \tag{107}$$

$$d_j = 1 - 2a_j - \frac{r\Delta t}{2}. \tag{108}$$

This is a tridiagonal system solved by LU decomposition at each time step. Boundary conditions: $V_0^n = e^{-r(n\Delta t)}H(0)$ and $V_M^n = e^{-r(n\Delta t)}H(1)$.

For American options, the backward induction step is modified to enforce the exercise constraint:

$$V_j^n \leftarrow \max\big(V_j^n,\ (K - P_j)^+\big) \tag{109}$$

at each time step after the implicit solve, implementing the linear complementarity problem (54) via a projected SOR (successive over-relaxation) method.

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